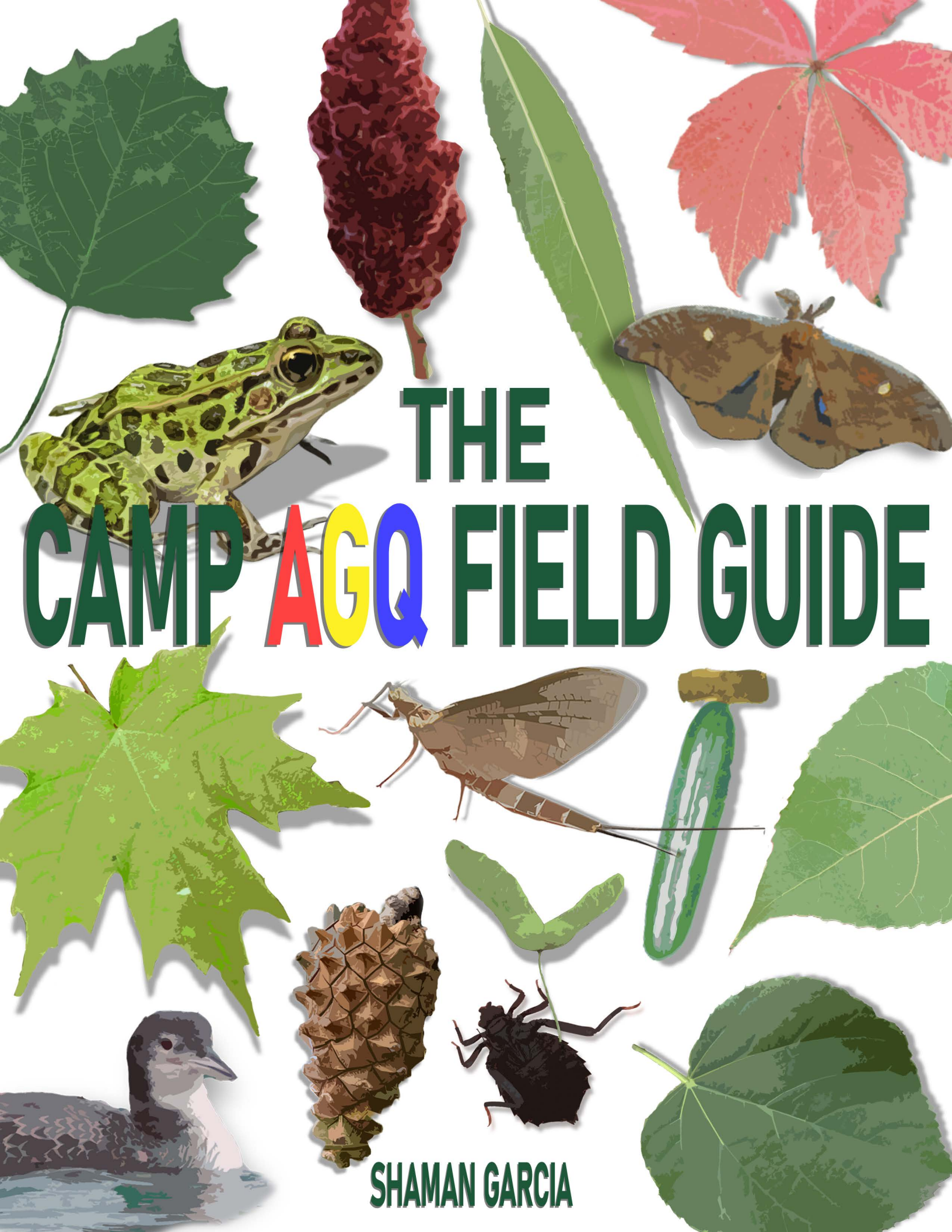




THE CAMP AGQ FIELD GUIDE

SHAMAN GARCIA

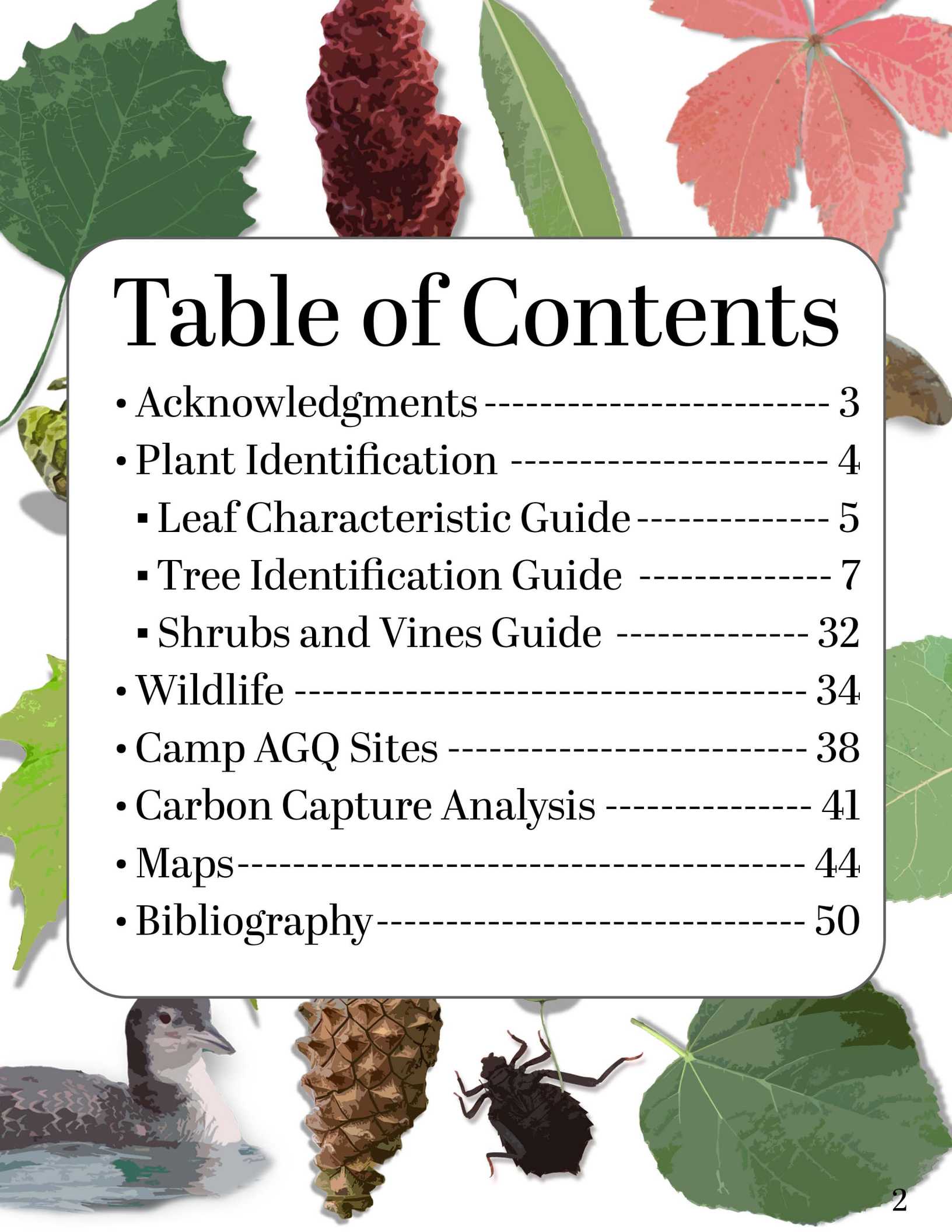


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Acknowledgments

I started attending Camp AGQ over 10 years ago. At first I was nervous, but it quickly became my favorite place. Since then, it is something I think about each day. My time at Camp AGQ taught me to explore, be curious, and appreciate nature and the environment around me. This field guide is a Capstone project as a culminating part of my Graduate degree in Environmental Science. I hope it serves to educate, inspire, and intrigue the ever-growing Camp AGQ family.



Plant Identification

The trees and plants around Camp AGQ are unique members of their ecosystem. Each shares similarities and differences with those around it, just like campers at AGQ. Learning how to identify the various native trees is a fun skill for anyone interested in growing their connection to nature around them.

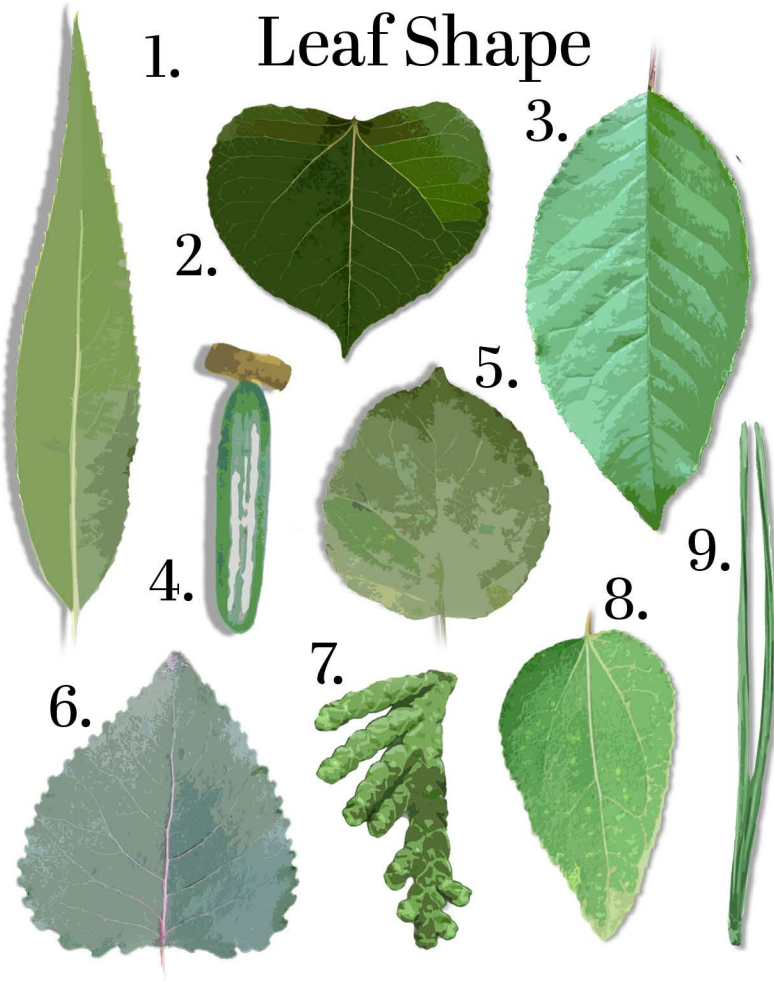
Plant Naming:

Common Name -- Eastern Cottonwood
(Scientific name) ----- (Populus deltoides)
Scientific Family ----- Salicaceae

- Common name is the local or regional name given to a species in English.
- Scientific name is the unique name given to a species in Latin. Names are given as Genus species.
- Scientific family is the name in Latin of the group in which a species belongs.

Leaf Characteristic Guide

1. Leaf Shape



- 1. Lanceolate
- 2. Cordate
- 3. Oval
- 4. Linear
- 5. Orbicular
- 6. Deltoid
- 7. Scale-like
- 8. Ovate
- 9. Needlelike

Leaf Tip



Leaf Base



Leaf Margin - (The perimeter of the leaf)



- Smooth**
(no teeth or serration)
- Serrate**
(zig-zag)
- Doubly Serrate**
(zig-zag within zig-zags)
- Toothed**
(distinct points)
- Lobed**
(leaf branches)

Leaf Characteristic Guide

Leaf Type

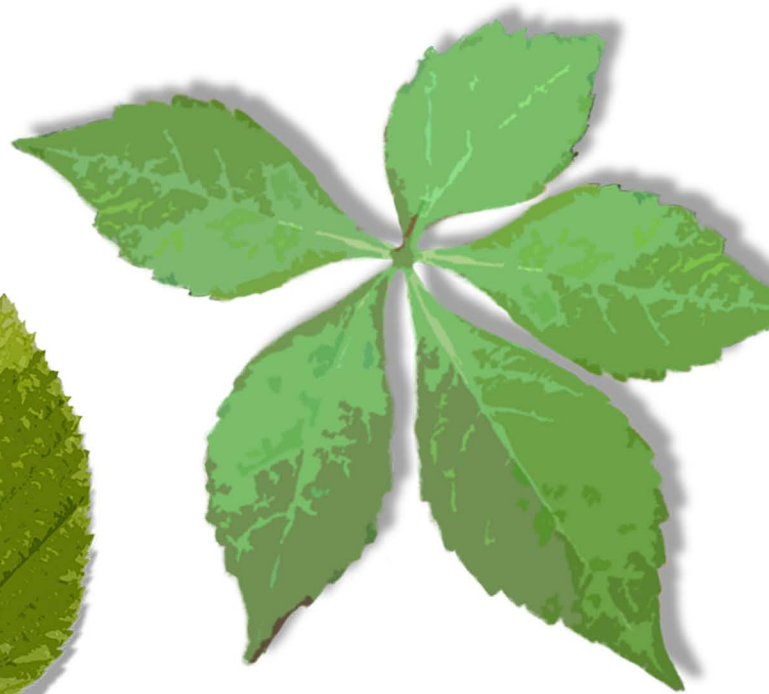
How single leaves or multiple leaflets connect to the stem.



Pinnately compound
Many leaflets growing along a stem



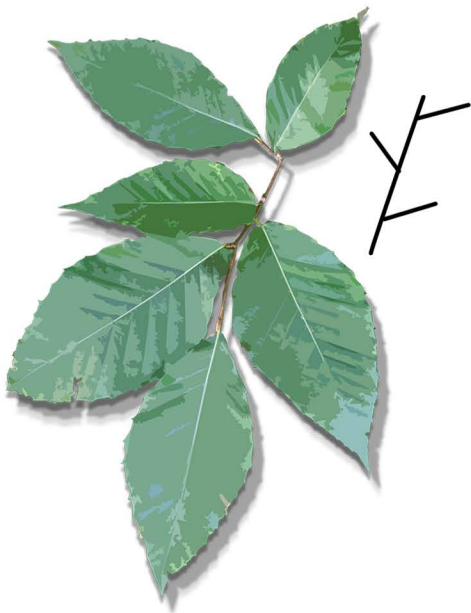
Simple
Single leaf attached to a single stem



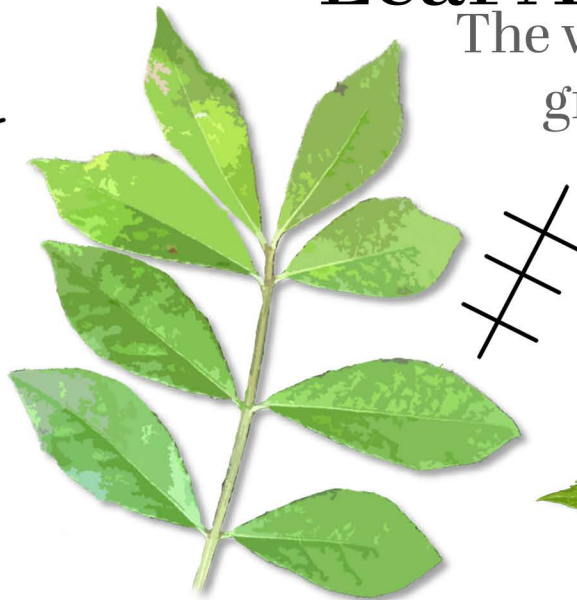
Palmately compound
Many leaflets growing from one spot

Leaf Arrangement

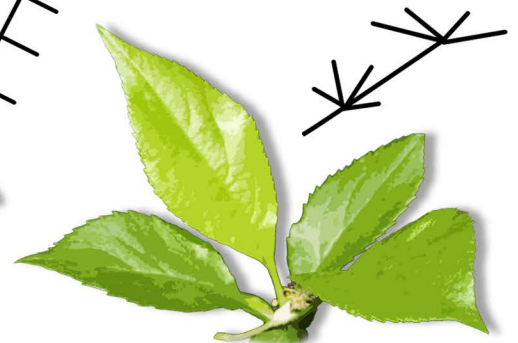
The way in which leaves grow along the twig.



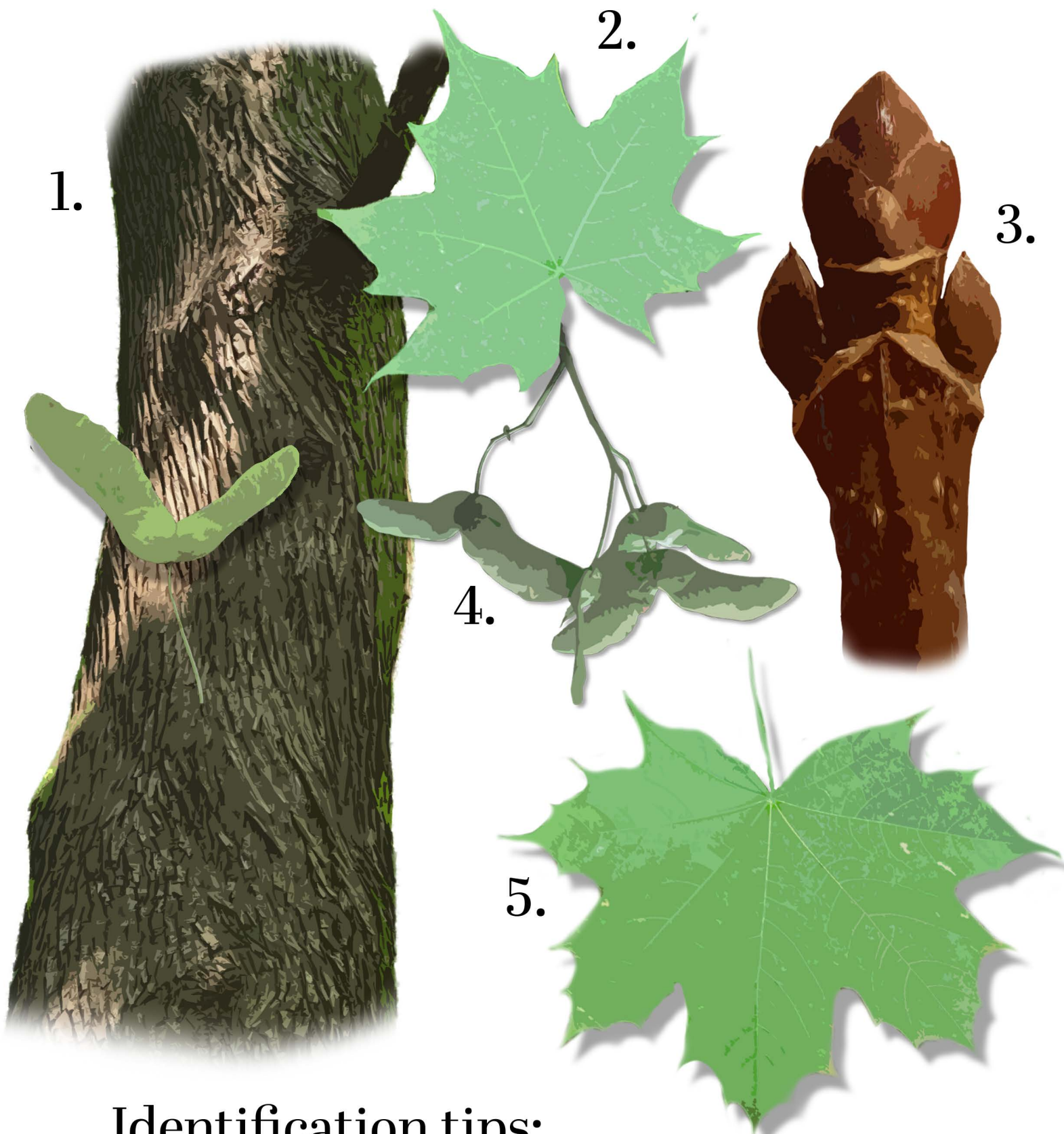
Alternate
Leaves are staggered along the twig



Opposite
Leaves grow across from one another



Whorled
Leaves spread out from one spot



Identification tips:

1. Bark is smooth with narrow ridges
2. Leaf is broad with 5 lobes
3. Bud is stout with few bud scales
4. Fruit is a paired divergent samara
5. Leaf margin is smooth with pointed tips

Norway Maple
(*Acer platanoides*)
Sapindaceae

Identification tips:

1. Bark is smooth and splotchy
2. Leaf is 3-5 lobed with a coarsely serrate margin
3. Fruit is a v-shaped samara
4. Fall colors are bright red
5. Bud is red with few bud scales

Red Maple (*Acer rubrum*)

Sapindaceae

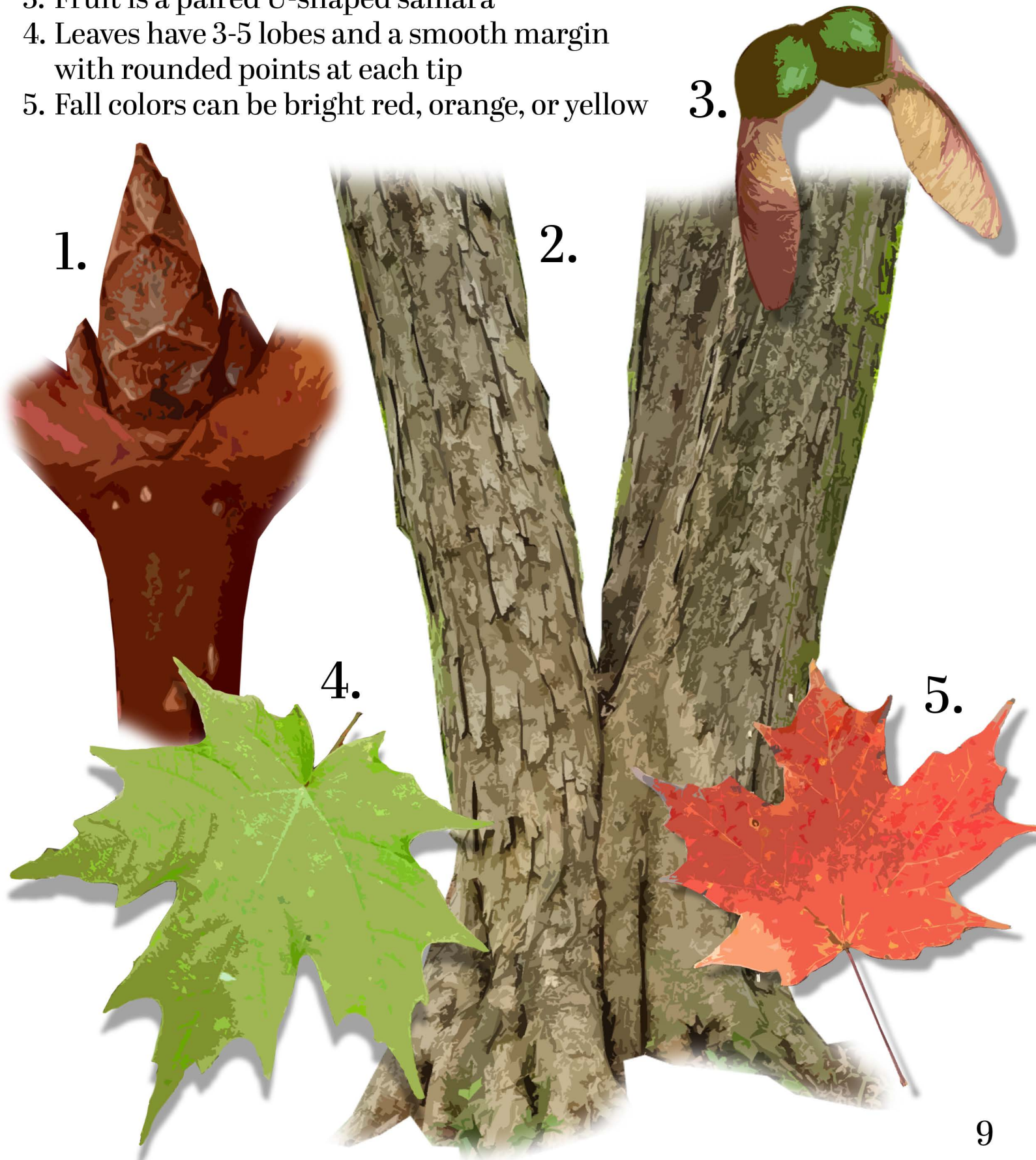


Identification tips:

1. Buds are pointed with visible bud scales
2. Barks is grey or light brown made up of overlapping irregular plates
3. Fruit is a paired U-shaped samara
4. Leaves have 3-5 lobes and a smooth margin with rounded points at each tip
5. Fall colors can be bright red, orange, or yellow

Sugar Maple (*Acer saccharum*)

Sapindaceae



Mountain Maple

(*Acer spicatum*)

Sapindaceae

Identification tips:

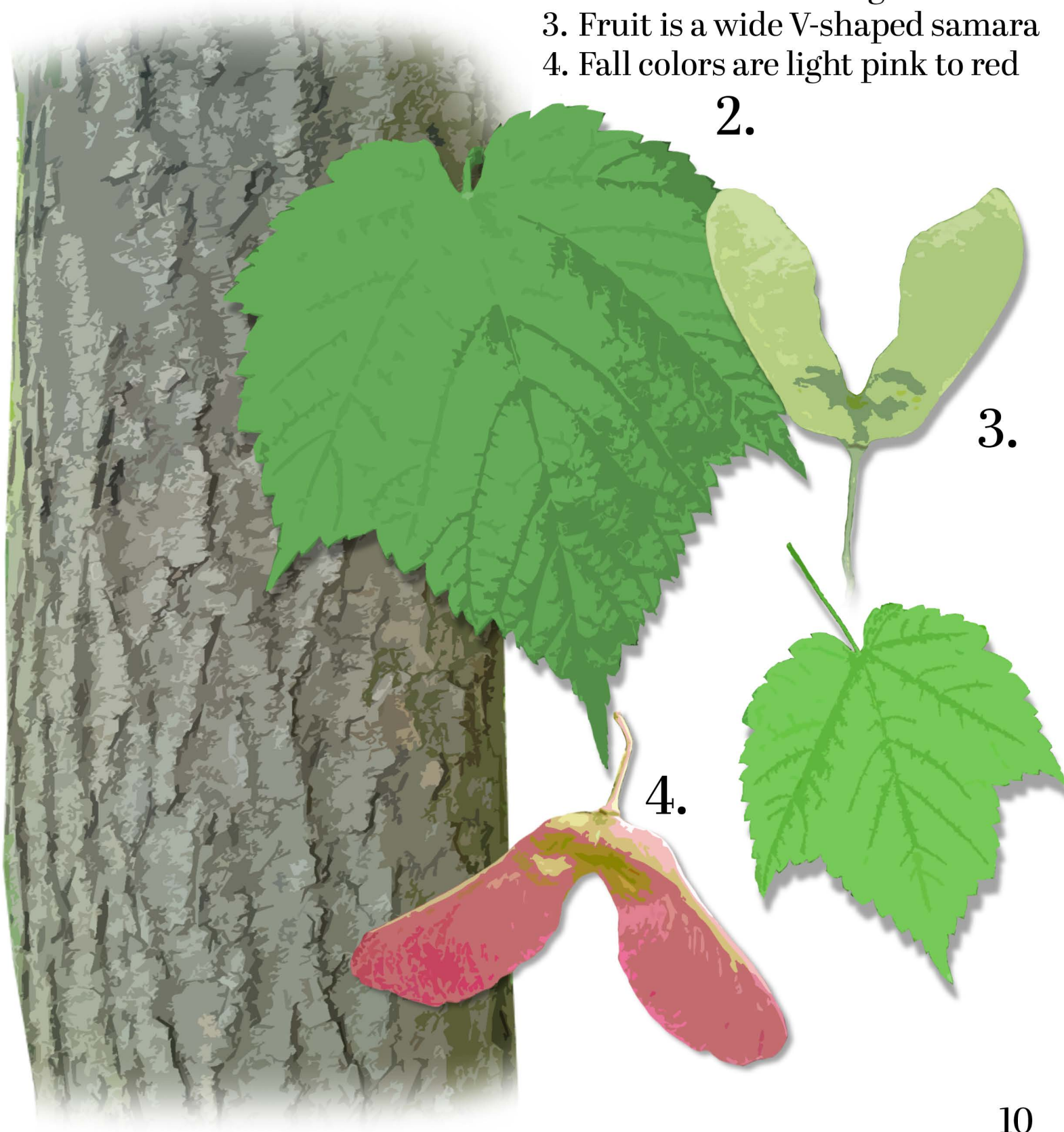
1. Young bark is smooth and gains irregular fissures as it matures
2. Leaf is cordate in shape with 3-5 wide lobes with a crenate-serrate margin
3. Fruit is a wide V-shaped samara
4. Fall colors are light pink to red

1.

2.

3.

4.

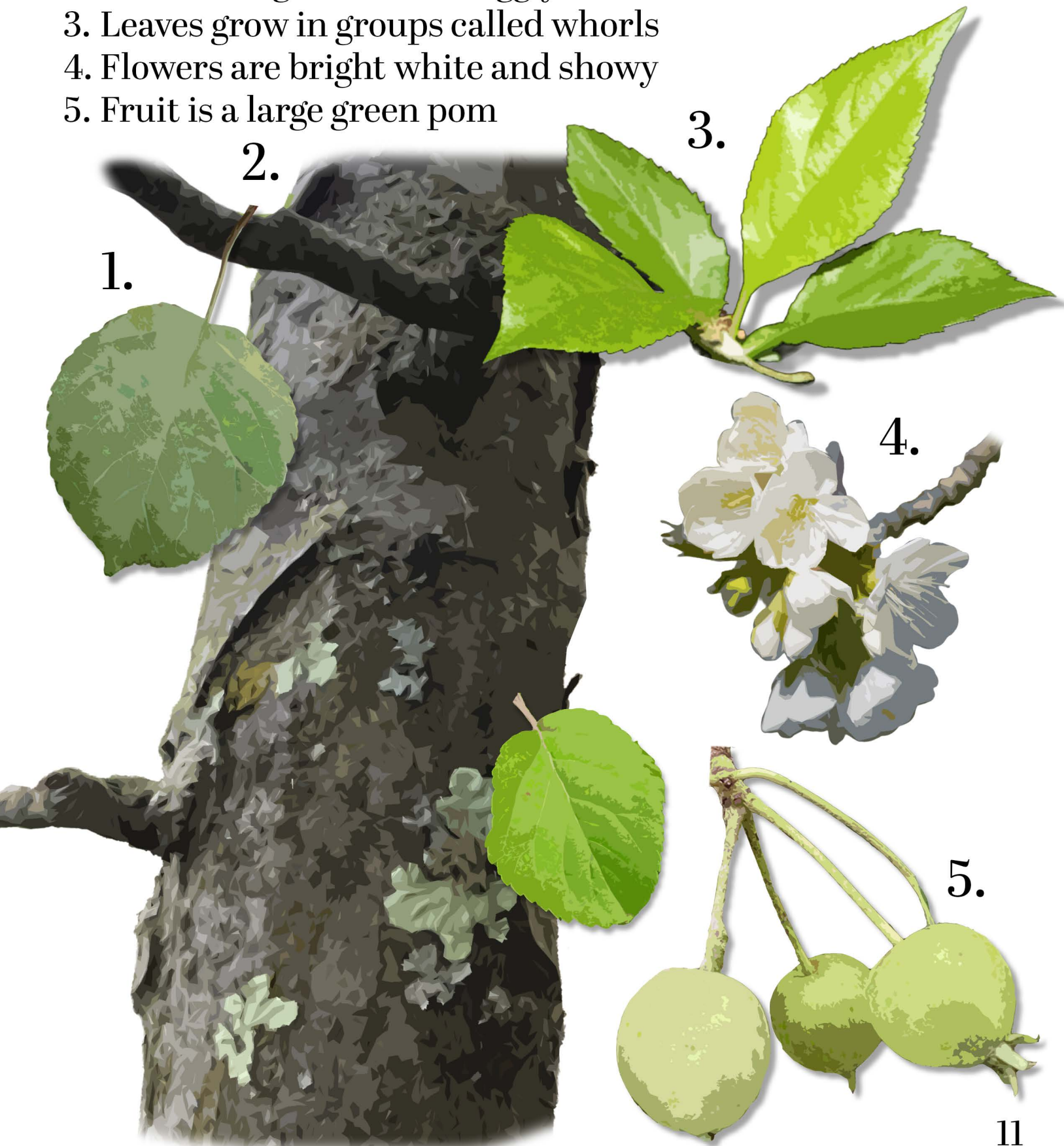


Identification tips:

1. Leaves vary from ovate to orbicular with a crenate-serrate margin
2. Bark is irregular and scraggly
3. Leaves grow in groups called whorls
4. Flowers are bright white and showy
5. Fruit is a large green pom

Crab Apple (*Malus* spp.)

Rosaceae

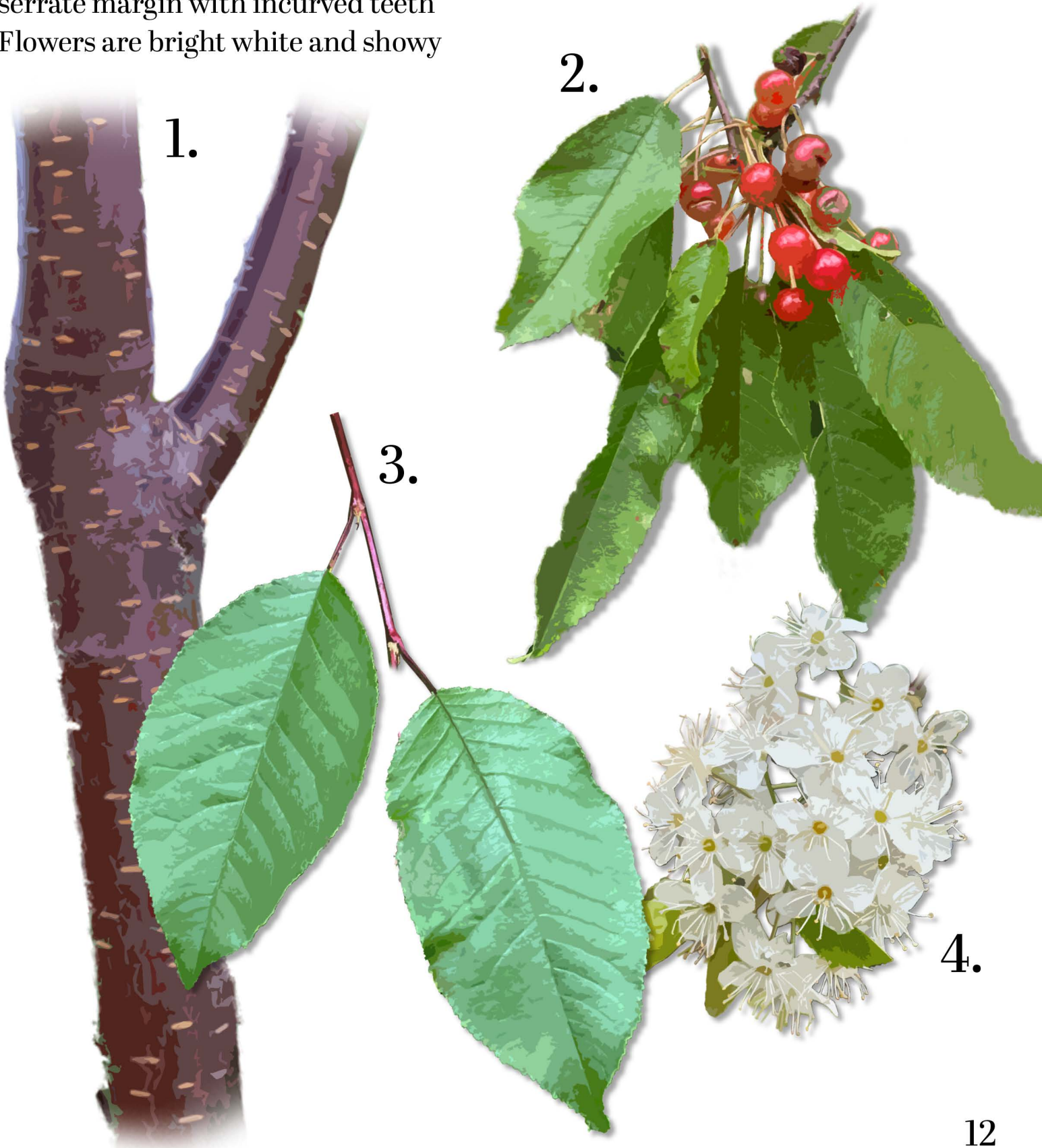


Identification tips:

1. Bark is smooth and red to brown with light brown lenticils
2. Fruit is a small red and shiny drupe
3. Leaf is lanceolate to obovate with a finely serrate margin with incurved teeth
4. Flowers are bright white and showy

Fire Cherry (*Prunus pennsylvanica*)

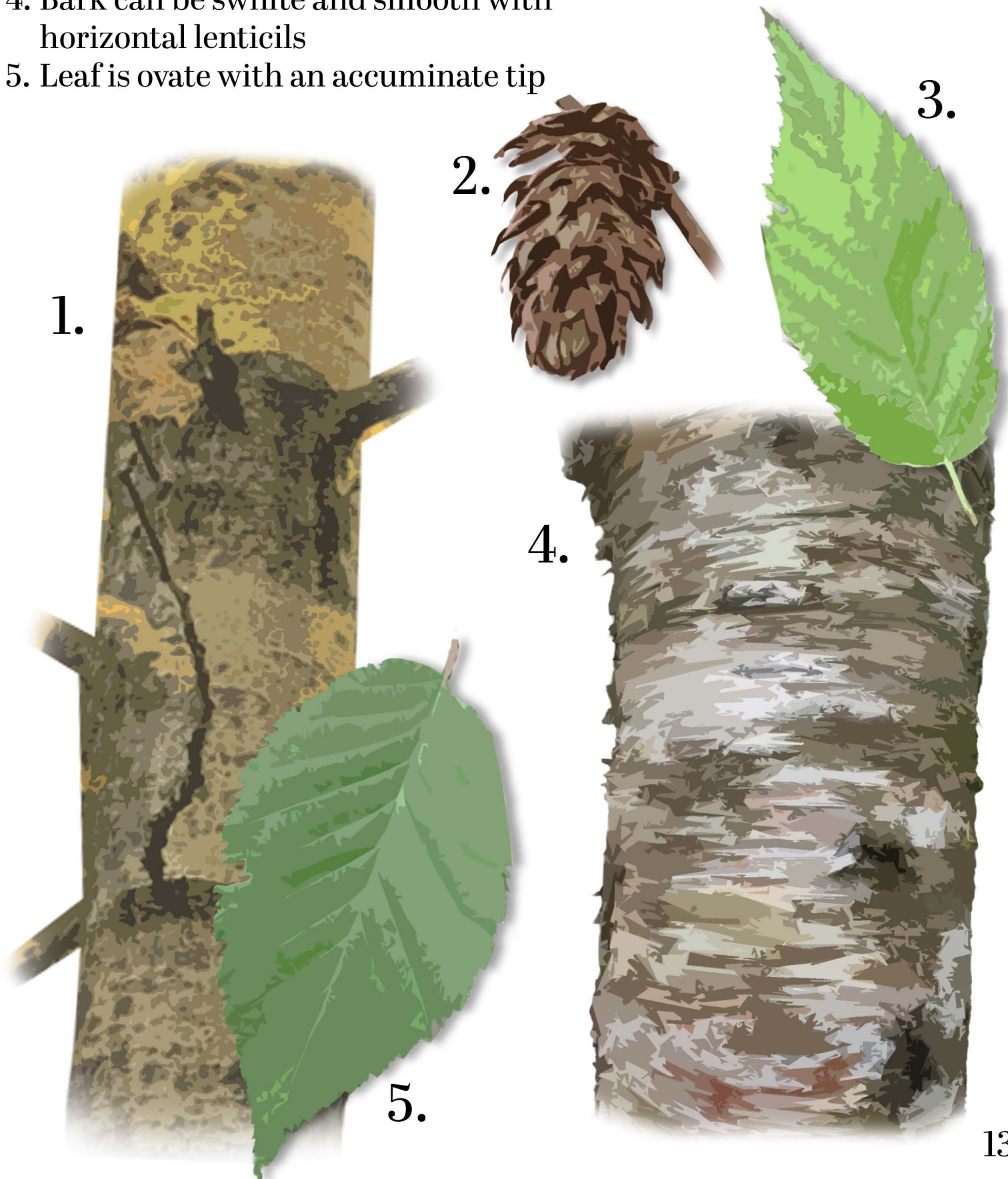
Rosaceae



Identification tips:

1. Bark is yellow to brown and smooth
2. Fruit is a cone-like catkin
3. Leaf margin is doubly serrate
4. Bark can be white and smooth with horizontal lenticils
5. Leaf is ovate with an acuminate tip

Yellow Birch (*Betula alleghaniensis*) Betulaceae



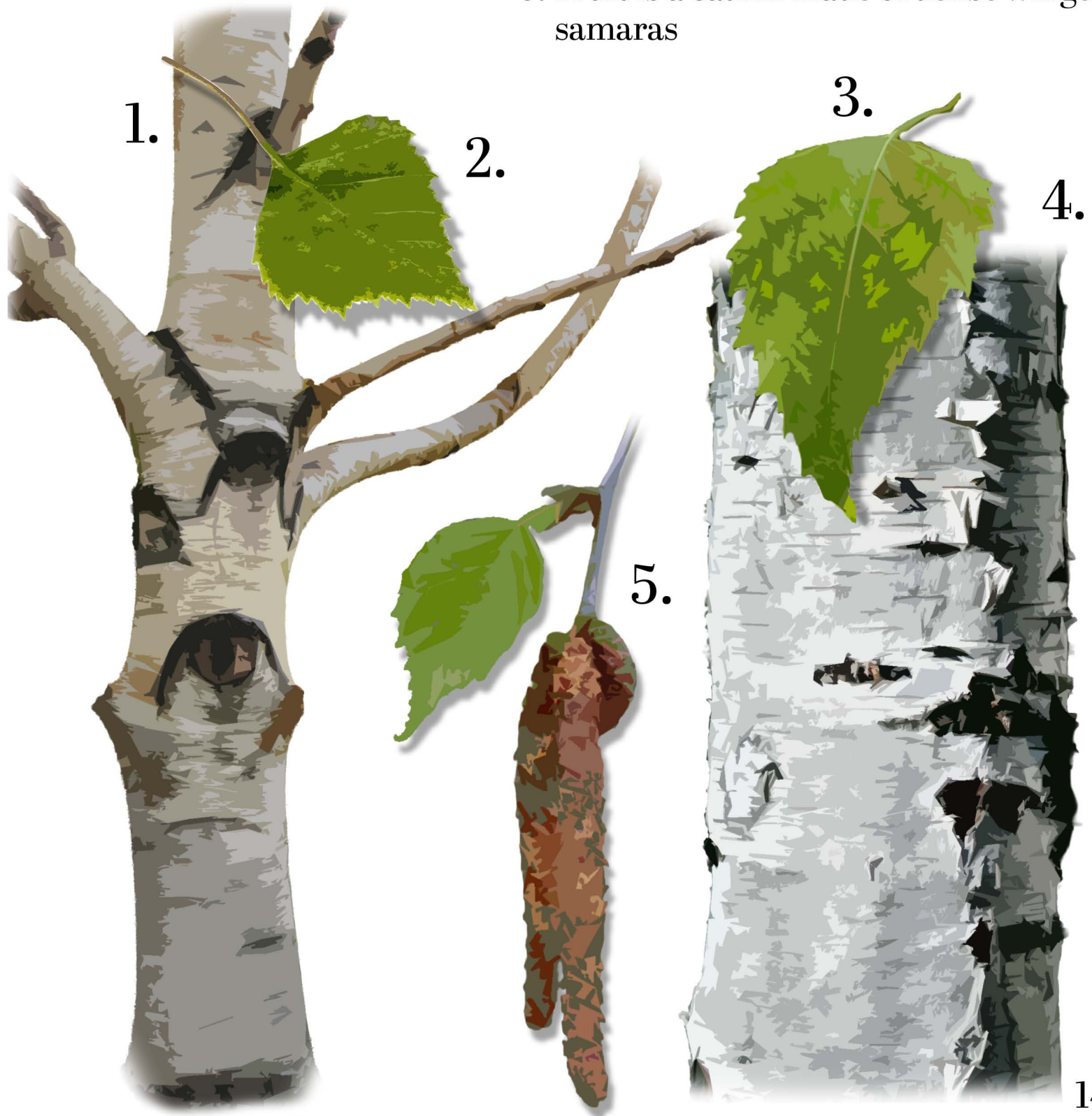
Paper Birch

(*Betula papyrifera*)

Betulaceae

Identification tips:

1. Tree is medium sized with smooth white bark
2. Leaf is ovate to deltoid
3. Leaf margin is coarsely doubly serrate
4. Bark peels up from the sides
5. Fruit is a catkin made of dense winged samaras

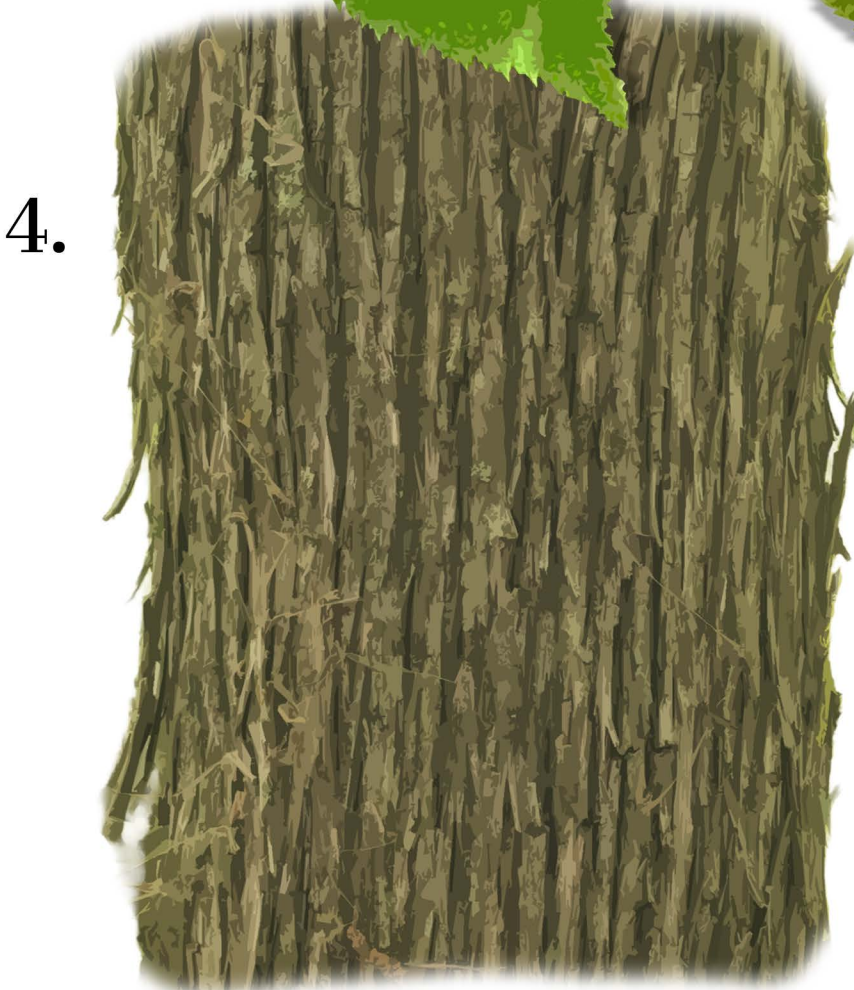


Ironwood

(*Ostrya Virginiana*)
Betulaceae

Identification tips:

1. Leaves and branches have alternate growth
2. Leaves are broadly oblong with a double serrate margin
3. Bark has thin vertical strips
4. Bark peels off on the edges

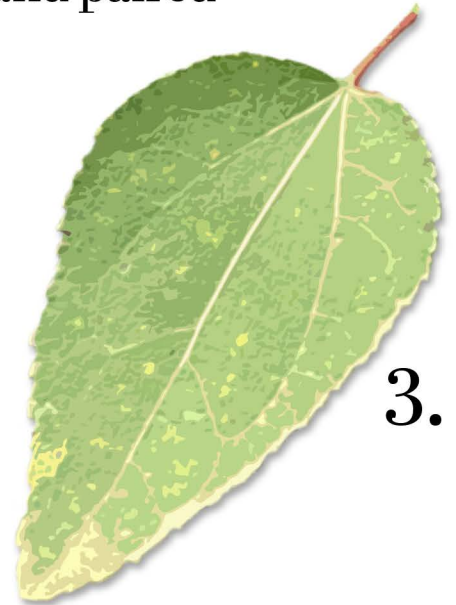


Balsam Poplar

(*Populus balsemifera*)
Salicaceae

Identification tips:

1. Leaf is obovate
2. Young tree showing opposite growth
3. Leaves have small rounded teeth along the margin
4. Buds are long and point inwards
5. Flower is fluffy and paired



Eastern Cottonwood

(*Populus deltoides*)

Salicaceae

Identification tips:

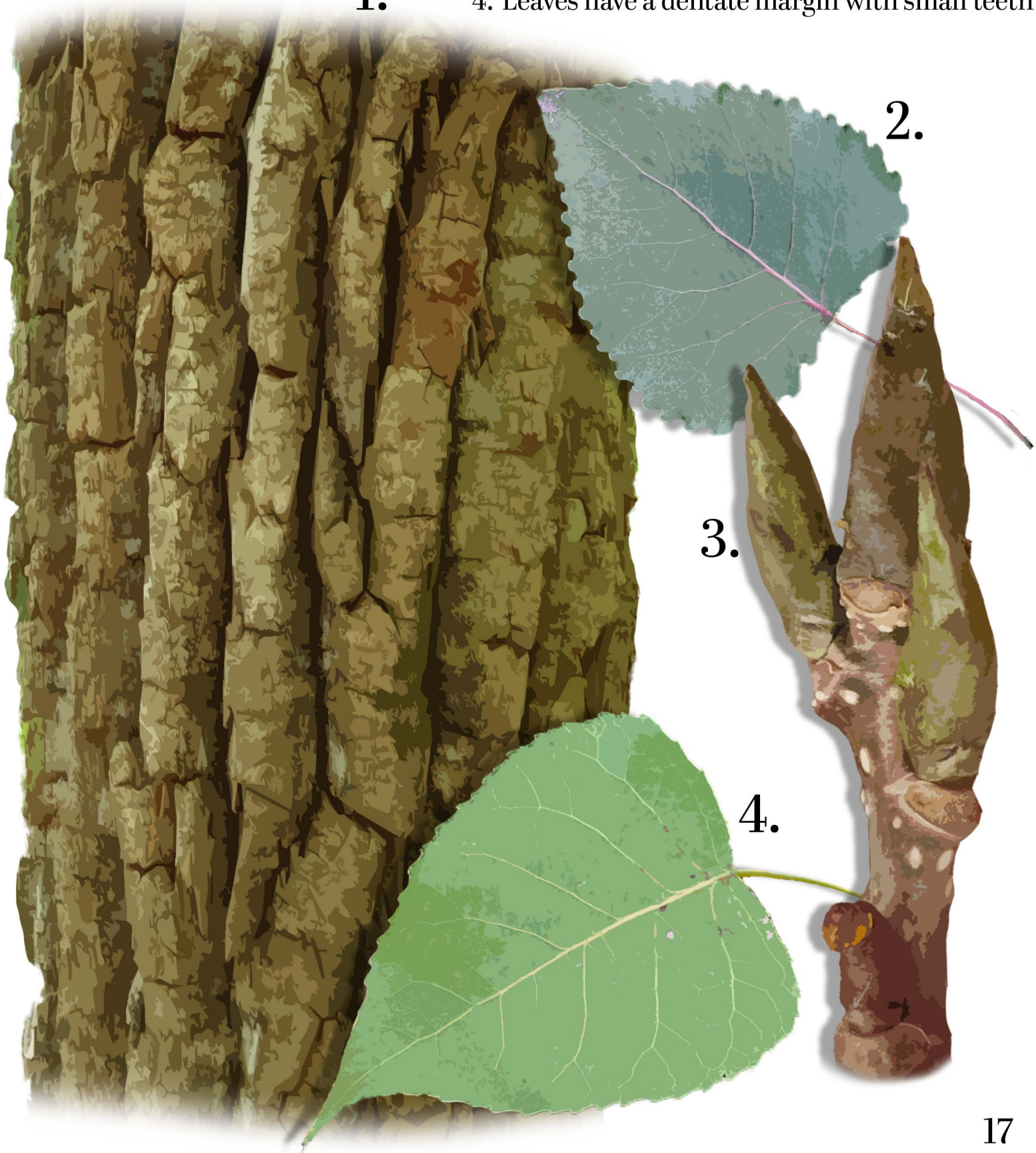
1. Bark has large ridges and deep fissures
2. Leaf is triangular in shape with a flat petiole
3. Buds are large, pointed, and sticky
4. Leaves have a dentate margin with small teeth

1.

2.

3.

4.



Bigtooth Aspen

(*Populus grandidentata*)
Salicaceae

Identification tips:

1. Young bark is smooth and white
2. Leaves are ovate to orbicular with large sinuate teeth
3. Mature bark is brown to gray with deep cracks and furrows

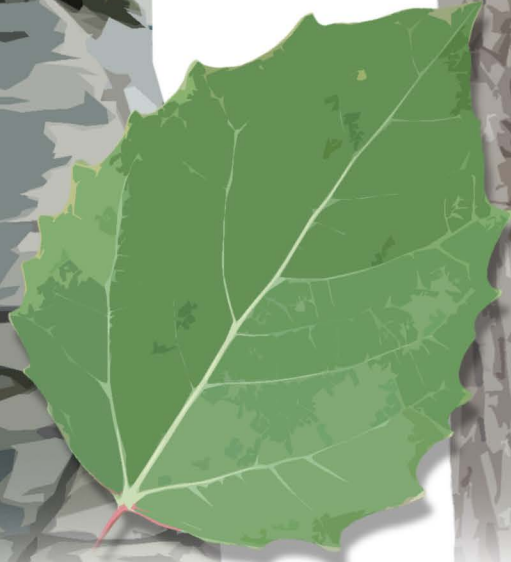
1.



3.



2.

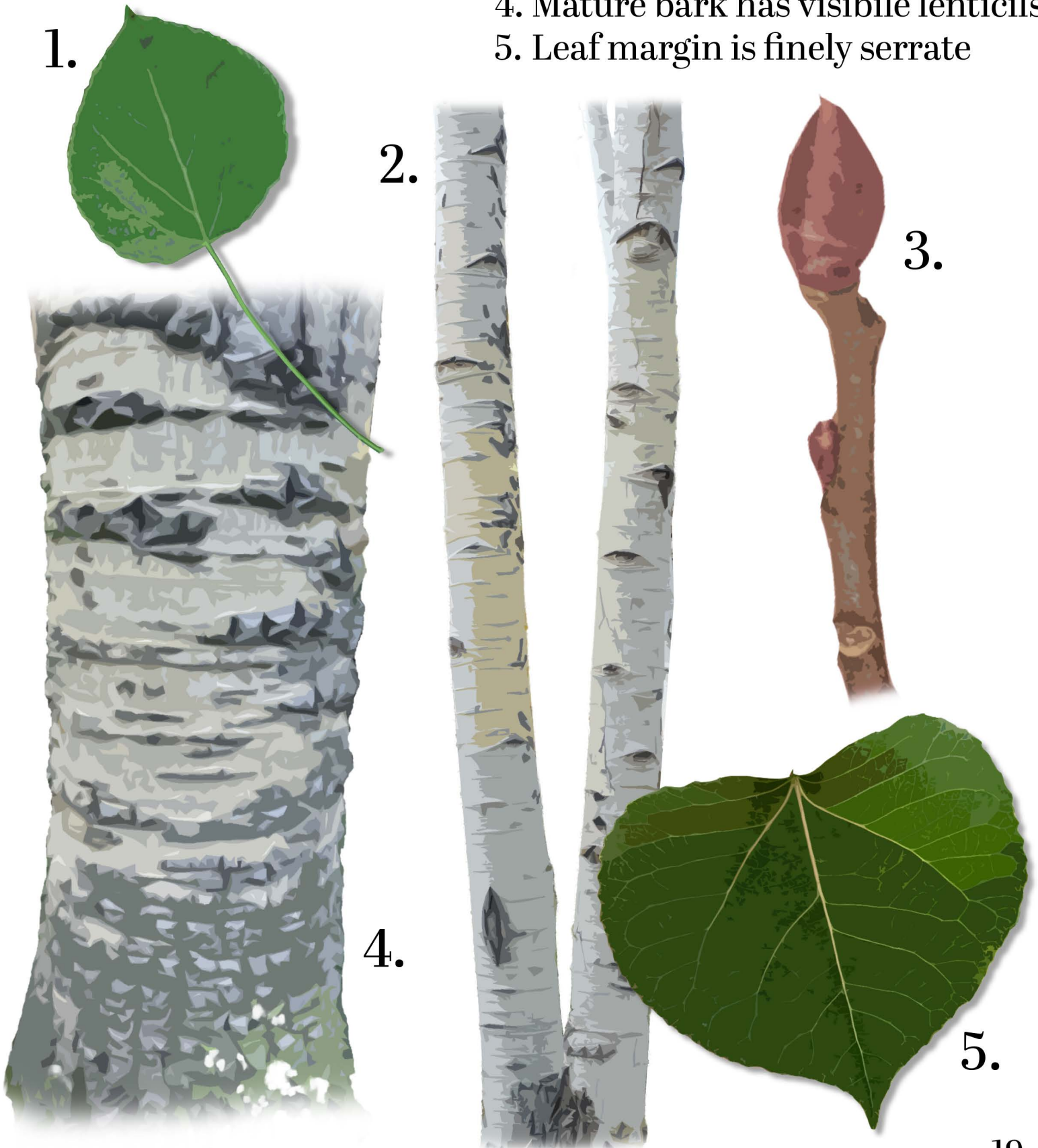


Trembling Aspen

(*Populus tremuloides*)
Salicaceae

Identification tips:

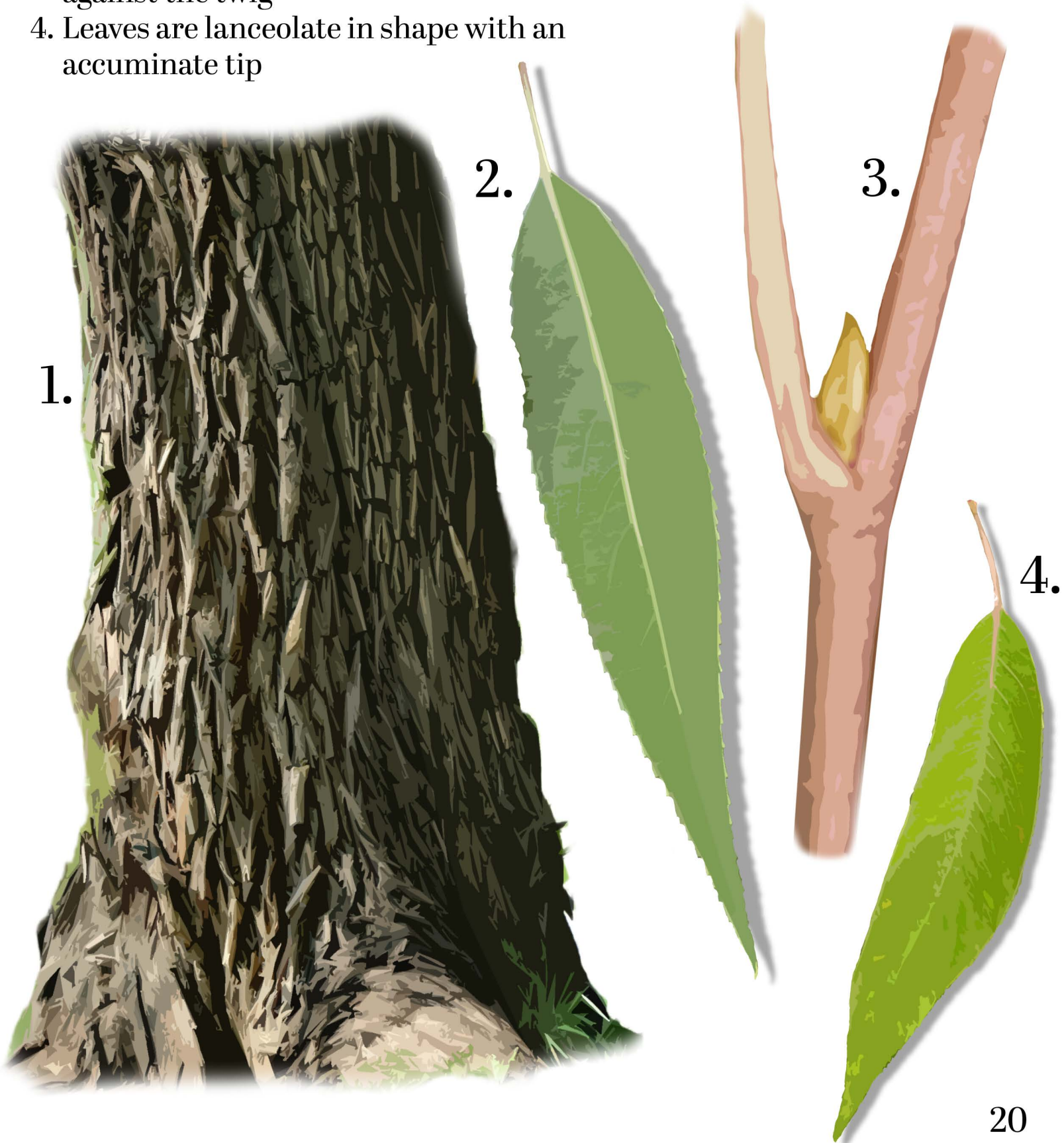
1. Leaves are cordate to ovate
2. Young bark is smooth and white
3. Buds are very large and pointed
4. Mature bark has visible lenticils
5. Leaf margin is finely serrate



Identification tips:

1. Bark has large raised irregular ridges
2. Leaves have a finely serrate margin
3. buds are light cream color and press against the twig
4. Leaves are lanceolate in shape with an acuminate tip

Peach-leaf Willow (*Salix amygdaloides*) Salicaceae

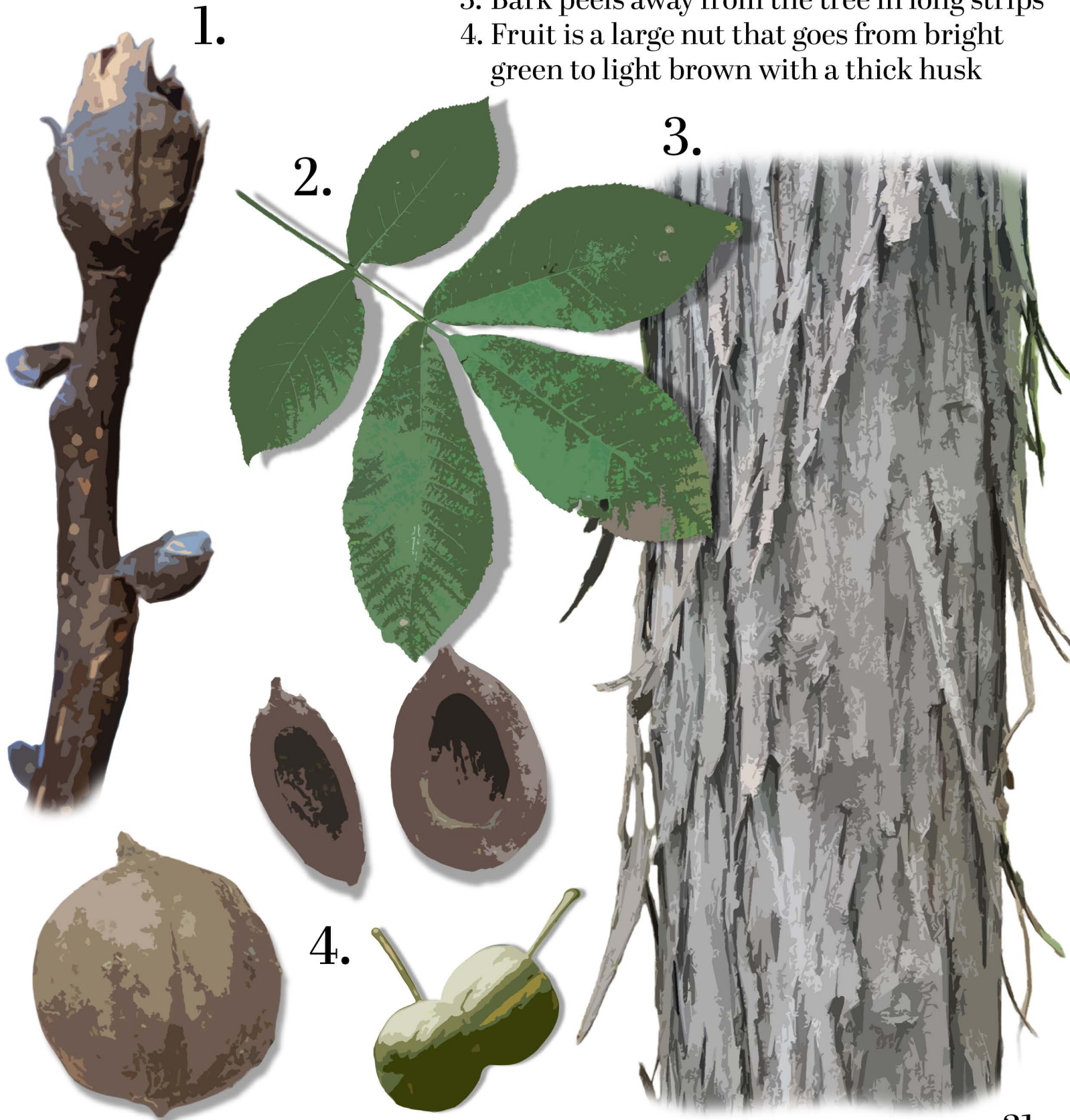


Shagbark Hickory

(*Carya ovata*)
Juglandaceae

Identification tips:

1. Bud is large and fuzzy
2. Leaf is made of 5-7 pinnately compound leaflets with a finely serrate margin
3. Bark peels away from the tree in long strips
4. Fruit is a large nut that goes from bright green to light brown with a thick husk



Identification tips:

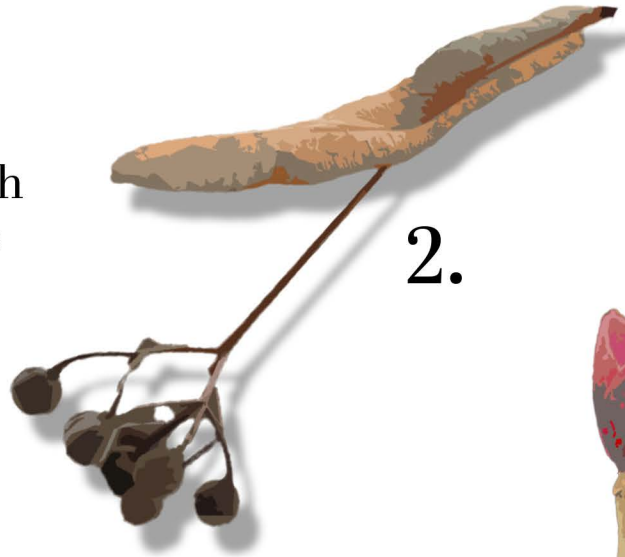
1. Bark is smooth with shallow interlacing X pattern
2. Fruit is a collection of small nuts attached to a wing-like modified leaf
3. Buds are large and egg shape and go from green to red in the fall
4. Leaf is large and cordate in shape with an inequilateral leaf base and serrate

American Basswood (*Tilia americana*) Malvaceae

1.



4.



2.



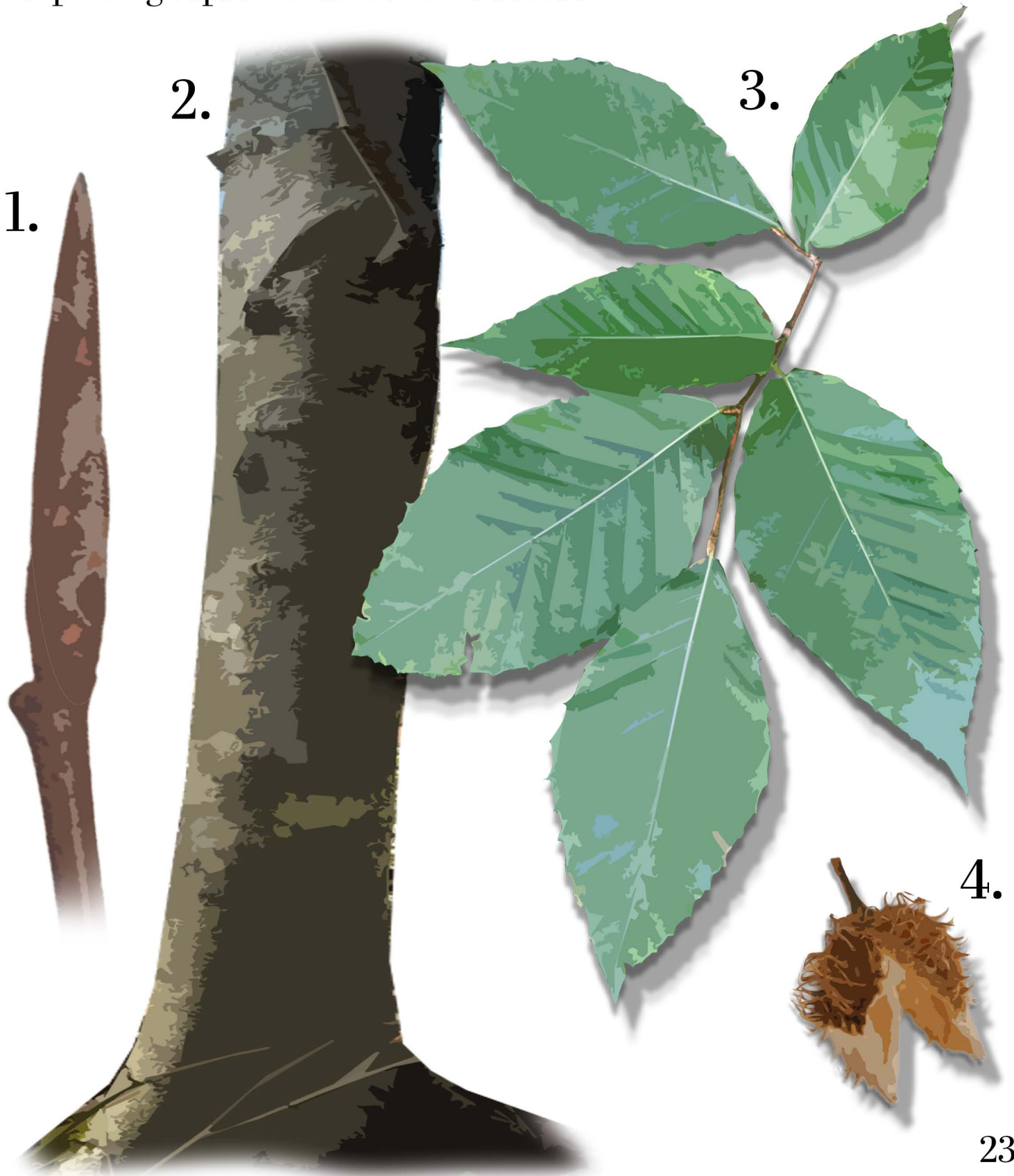
3.

5.

Identification tips:

1. Buds are long slender and brown
2. Bark is smooth and gray
3. Leaves are obovate with coarse teeth
4. Fruit is a nut or “Beech nut” which is an exploding capsule that launches seeds!

American Beech (*Fagus grandifolia*) Fagaceae



Pin Oak

(*Quercus ellipsoidales*)
Fagaceae

Identification tips:

1. Leaf has bristle tipped lobes with large sinuses
2. Form is a large bushy tree
3. Fruit is an acorn with a large cap and vertical stripes on the nut
4. Bark is flattened with gray streams



4.

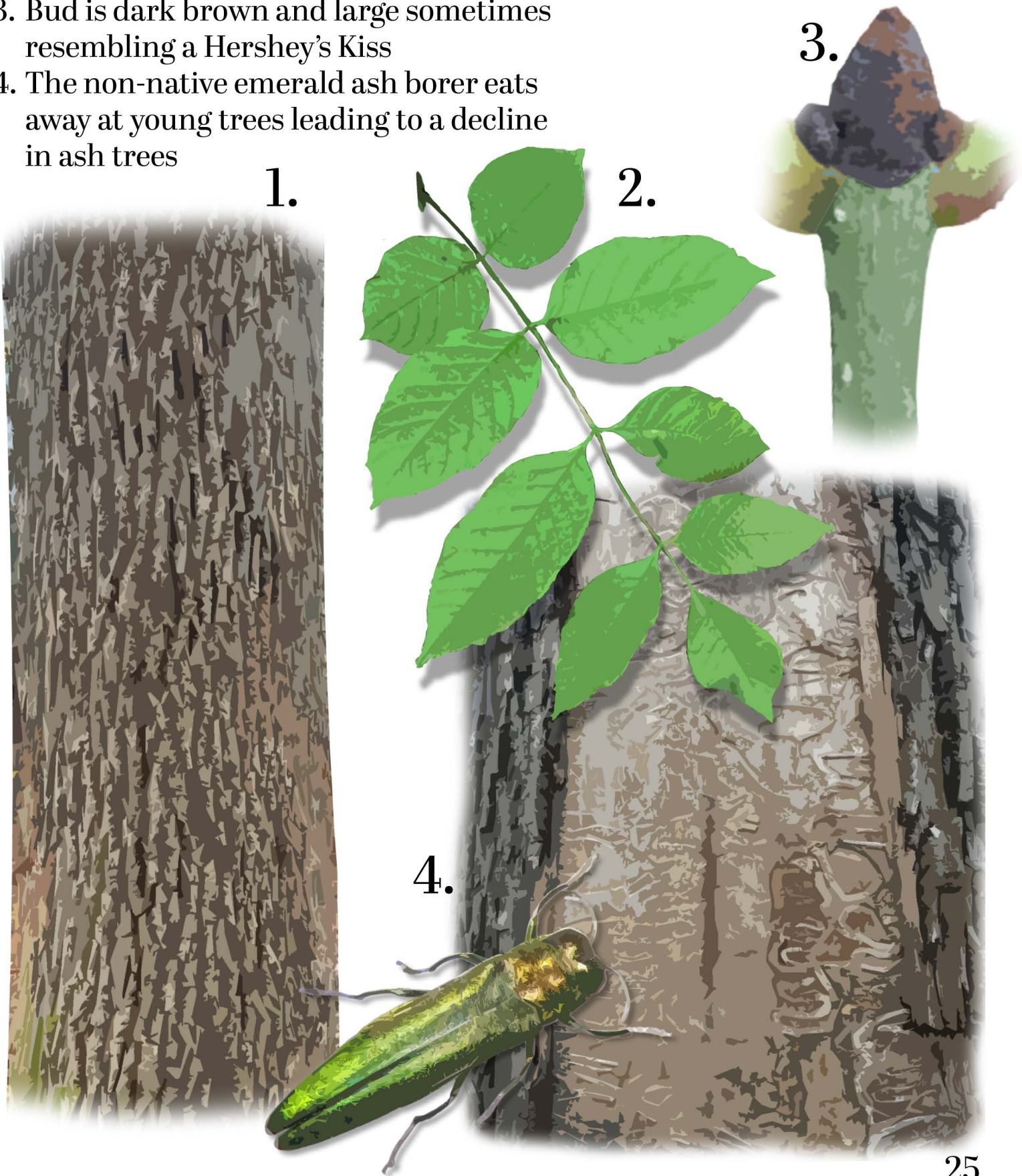


Identification tips:

1. Bark has raised ridges moving upwards
2. Leaf is made of 5-9 oblong to lanceolate leaflets with a smooth margin
3. Bud is dark brown and large sometimes resembling a Hershey's Kiss
4. The non-native emerald ash borer eats away at young trees leading to a decline in ash trees

White Ash (*Fraxinus americana*)

Oleaceae



Red/Green Ash

(*Fraxinus pennsylvanica*)

Oleaceae

Identification tips:

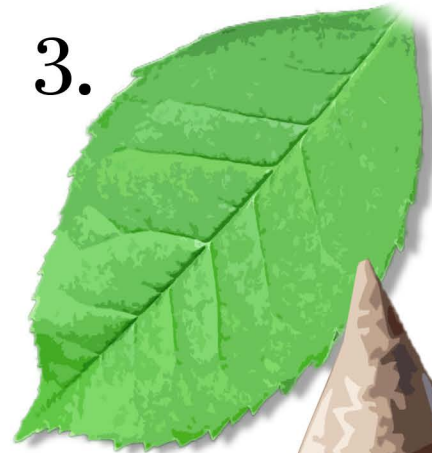
1. Young bark is light brown and warty
2. Leaf is made of 5-9 pinnately compound leaflets
3. Leaflets have a coarsely serrate margin
4. Fruits are samaras in dense clusters
5. Buds are large and brown and triangular with a gap between the terminal bud and lateral buds

1.



2.

3.



4.



5.



White-cedar (*Thuja occidentalis*) Cupressaceae

Identification tips:

1. Leaves are flattened with many scales
2. Bark is shreddy with long vertical stripes
3. Mature cones are small and brown
4. Young cones are light green and clustered



Eastern Hemlock

(*Tsuga canadensis*)
Pinaceae

Identification tips:

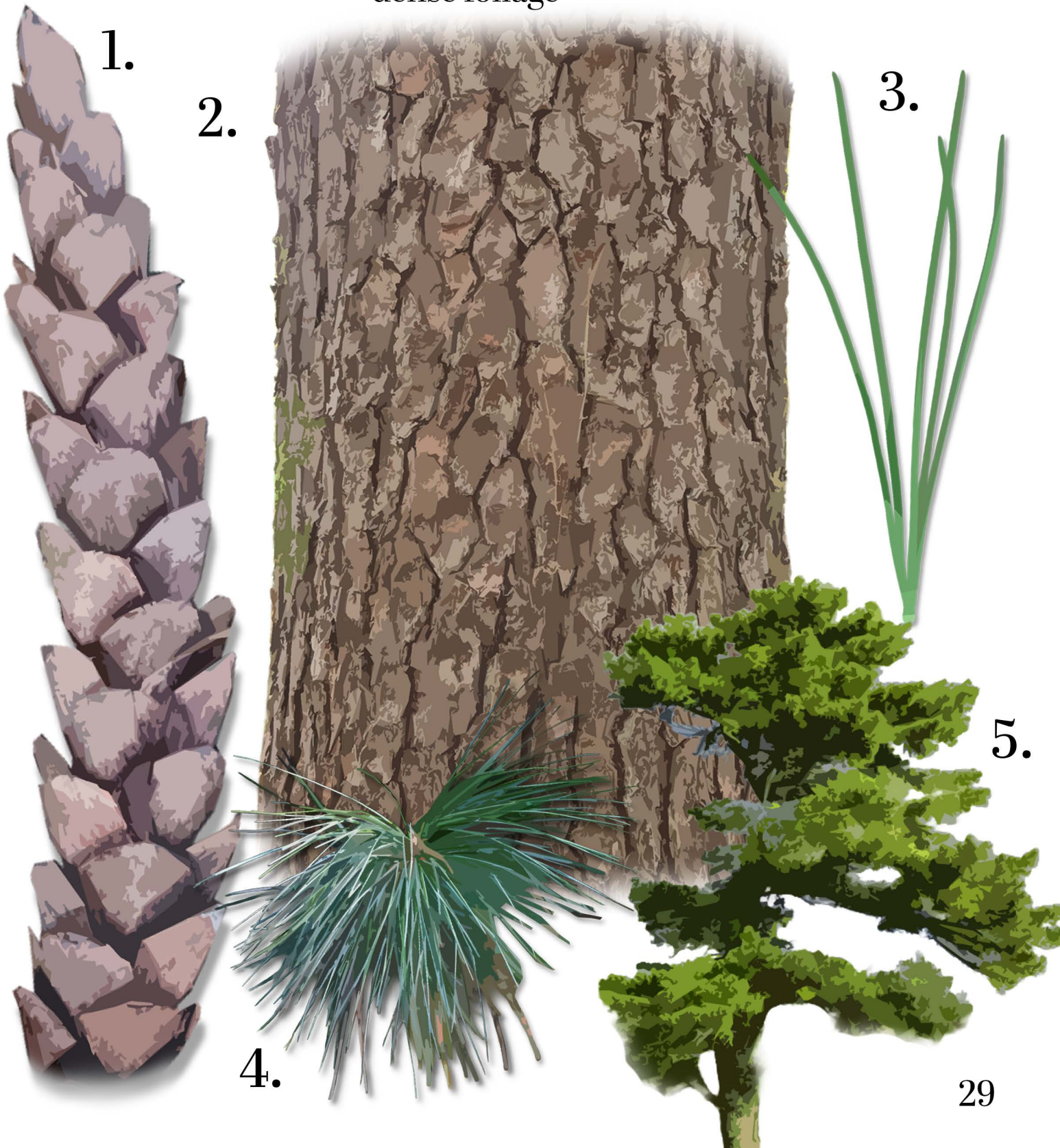
1. Needle-like leaves spiral around the twig
2. Bark is brownish gray and scaly with irregular fissures
3. Leaves have two white stripes on bottom side
4. Cone is light brown and cone shaped



Identification tips:

White Pine (*Pinus strobus*) Pinaceae

1. Long slender cone
2. Cracked mud bark
3. Long straight needles in groups of five
4. Fluffy paintbrush-like collections of needles
5. Excurrent growth with sparse branches and dense foliage



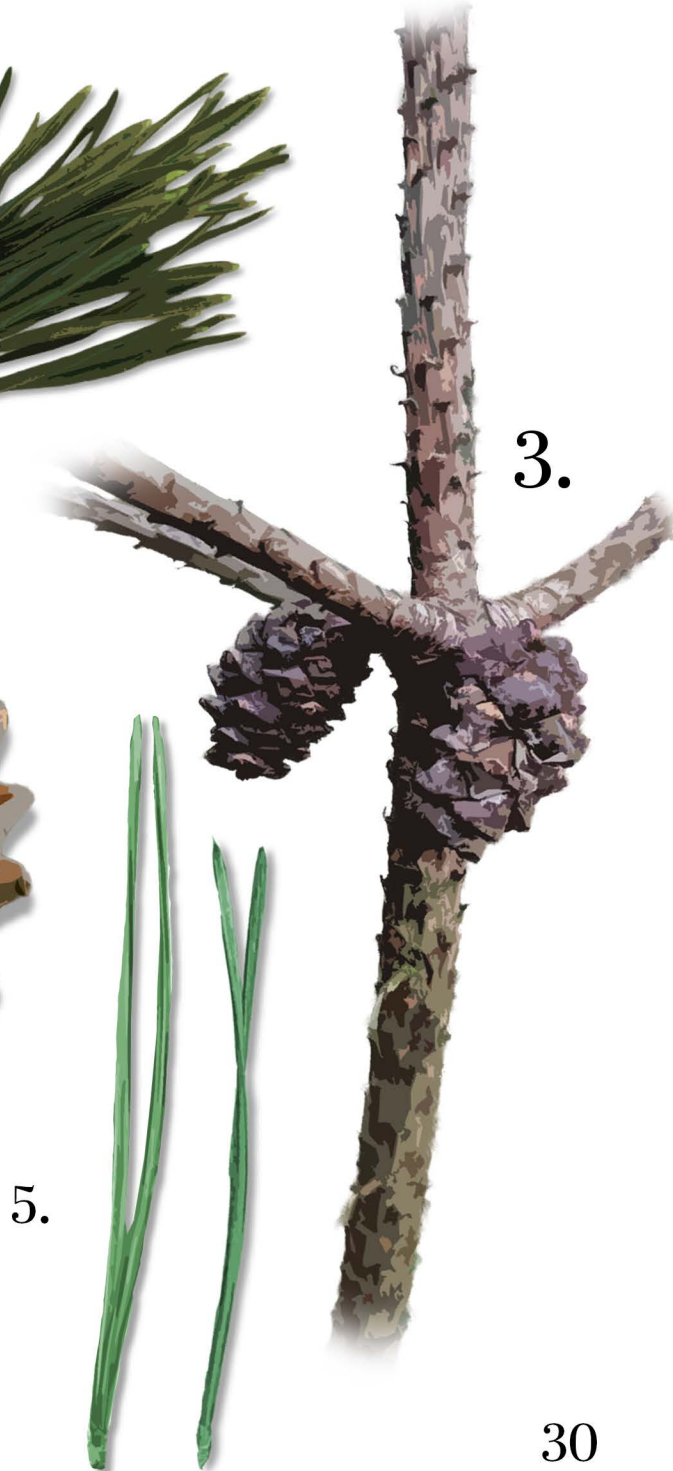
Scott's Pine

(*Pinus sylvestris*)

Pinaceae

Identification tips:

1. Bark goes from brown and scaly to orange and flaky
2. Leaves grow in needle-like clusters
3. Bare twig has peeling scales
4. Cone has volcano shaped scales
5. Needles grow in pairs that spiral

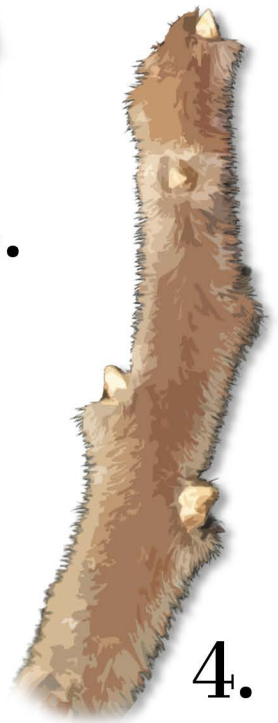


Staghorn Sumac

(*Rhus typhina*)
Anacardiaceae

Identification tips:

1. Leaves are pinnately compound with 10-30 leaflets
2. Leaflets are lanceolate with a sharply serrate margin
3. Fruit is a collection of fuzzy berry-like drupes
4. Twig and bark have fine dense hairs
5. Form is a small to medium showy shrub



Poison Ivy

(*Toxicodendron radicans*)

Anacardiaceae



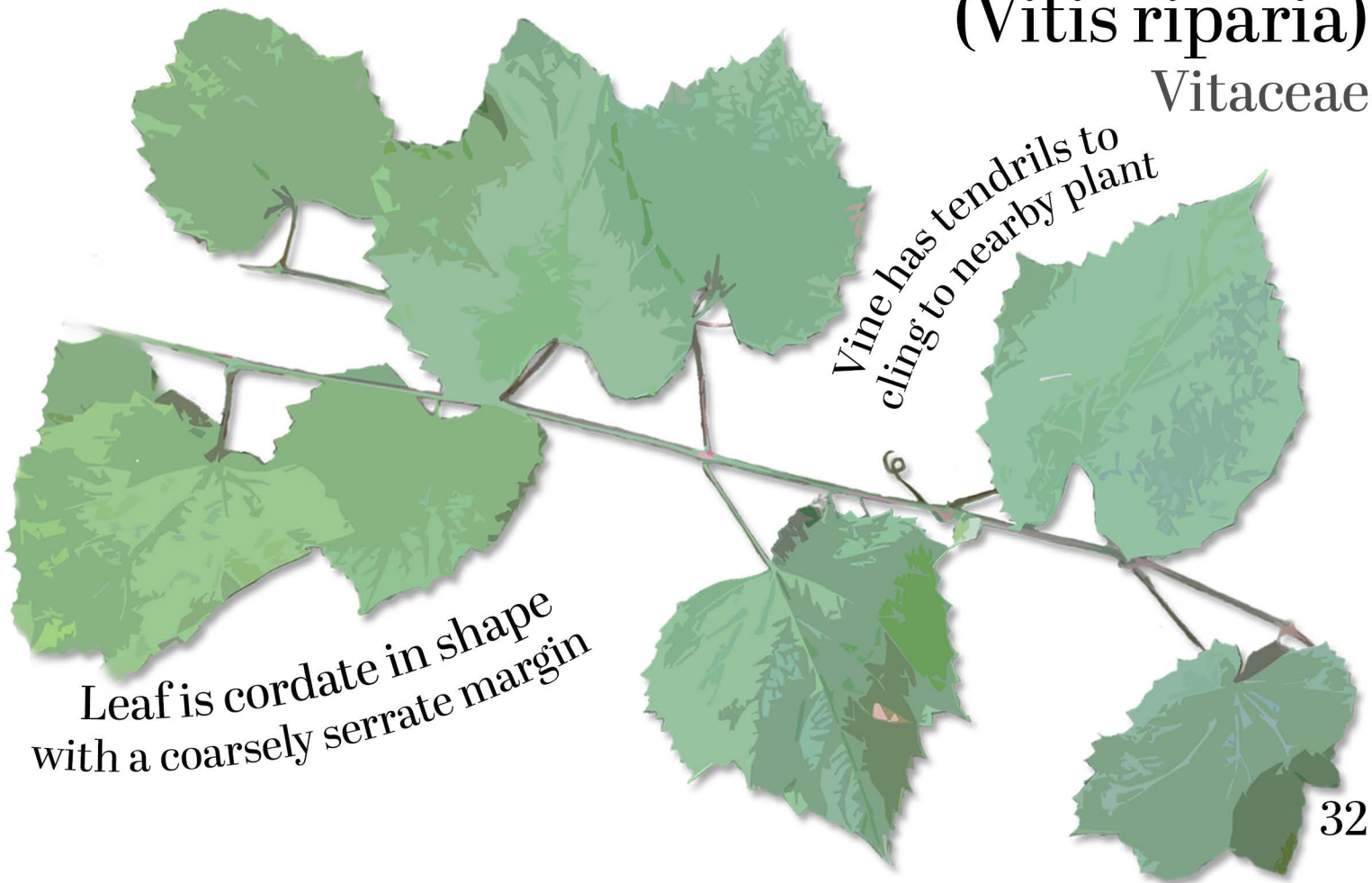
Leaflets of three

Leaflets vary in
size and shape

Riverbank-Grape

(*Vitis riparia*)

Vitaceae



Leaf is cordate in shape
with a coarsely serrate margin

Vine has tendrils to
cling to nearby plant

Virginia Creeper (*Parthenocissus quinquefolia*)

Vitaceae



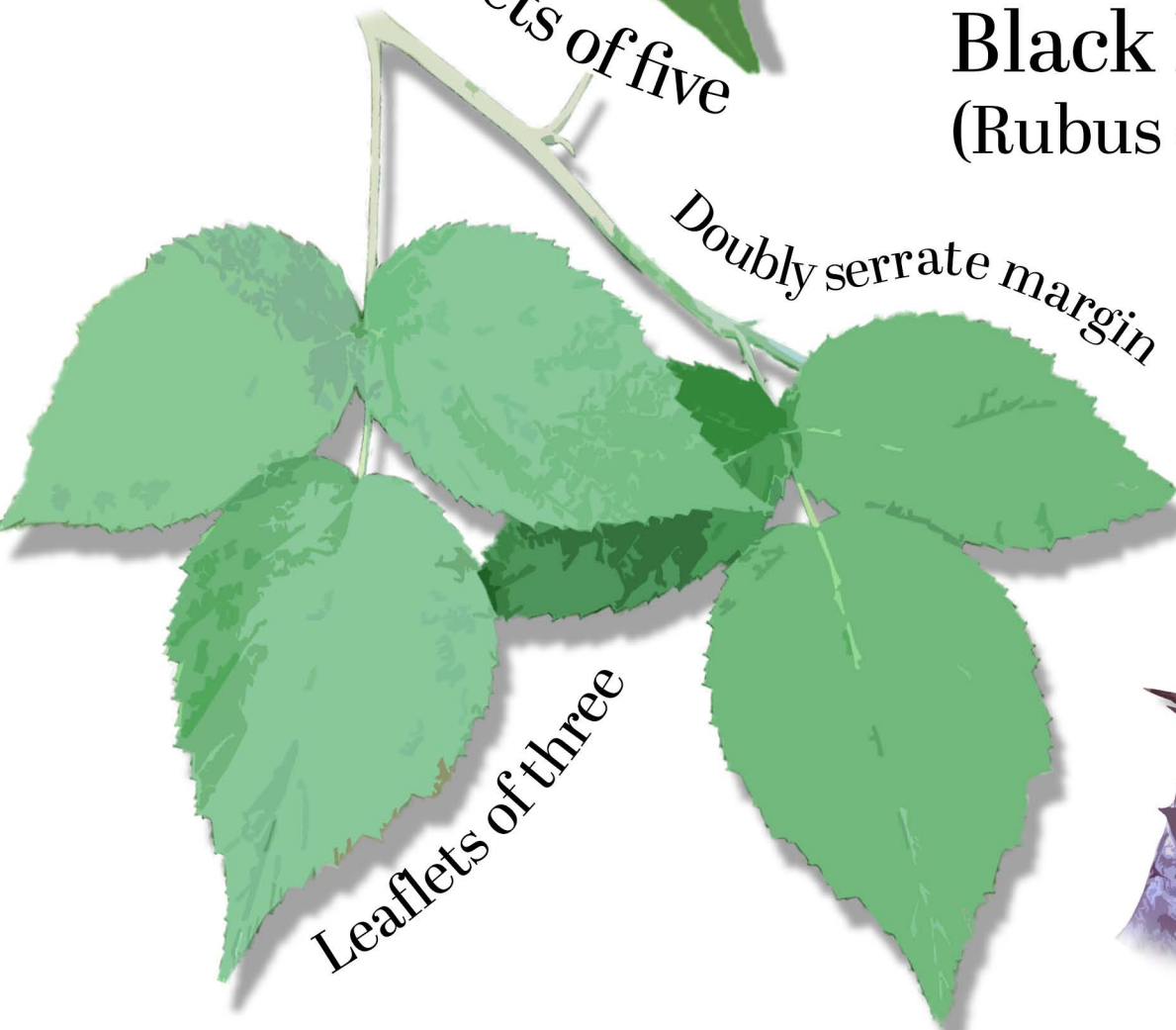
Leaflets of five



Pink or red fall colors
Coarsely serrate or
dentate margin

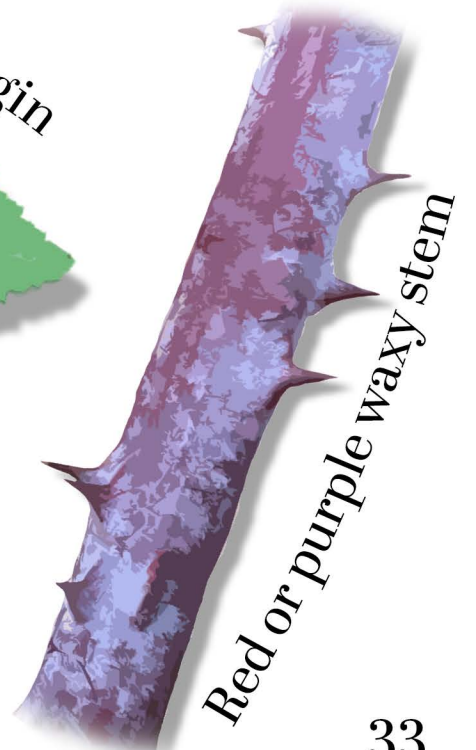
Black Raspberry (*Rubus occidentalis*)

Rosaceae



Doubly serrate margin

Leaflets of three



Red or purple waxy stem



Wildlife



Amphibians

Leopard Frog
(*Lithobates pipens*)



American Bullfrog
(*Lithobates catesbeiana*)



Eastern Red-Backed Salamander
(*Plethodon cinereus*)



Wood Frog
(*Lithobates sylvaticus*)



Green Frog
(*Lithobates clamitans*)



American Toad
(*Anaxyrus americanus*)

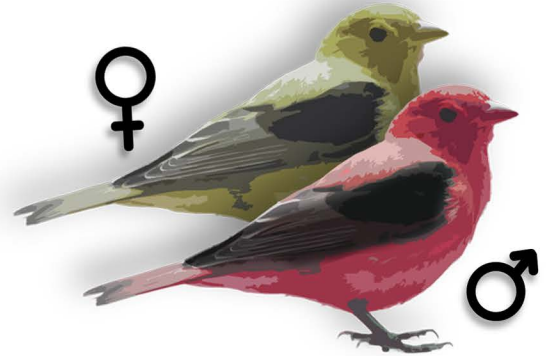


Birds

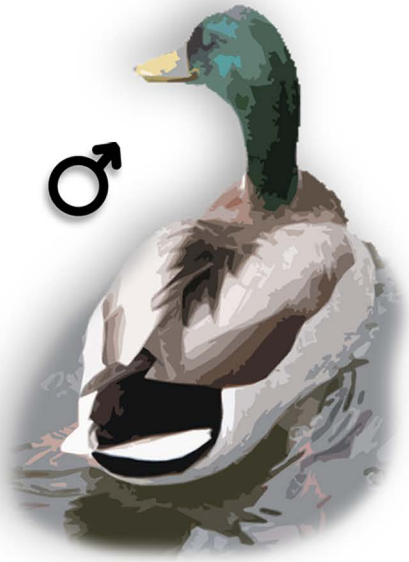


Common Loon
(*Gavia immer*)

Scarlet Tanager
(*Piranga olivacea*)



Mallard
(*Anas platyrhynchos*)



Hairy Woodpecker
(*Leuconotopicus villosus*)

Bald Eagle
(*Haliaeetus leucocephalus*)



Downy Woodpecker
(*Picoides pubescens*)



Ring-billed Gull
(*Larus delawarensis*)



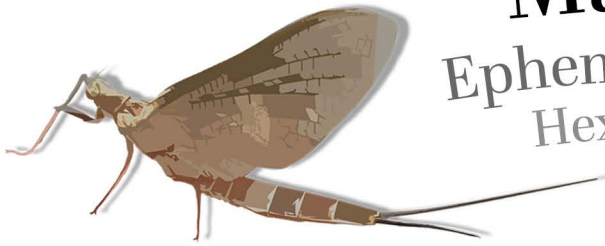
Insects



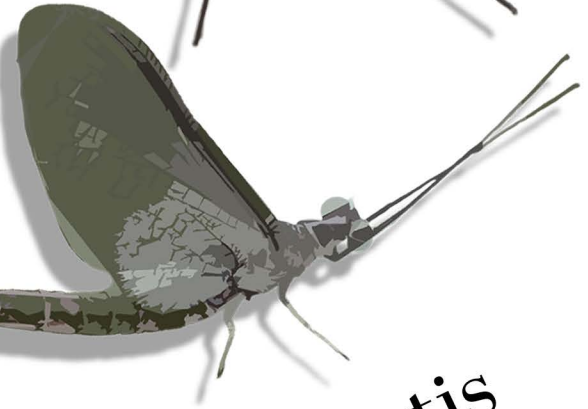
Mosquito
Culicidae



♀



Mayfly
Ephemeroptera
Hexegenia

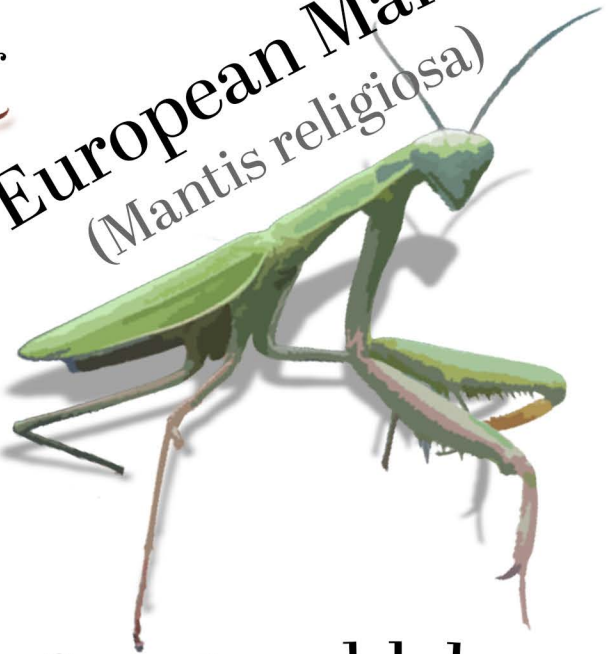


Polyphemus Moth
(*Antheraea polyphemus*)

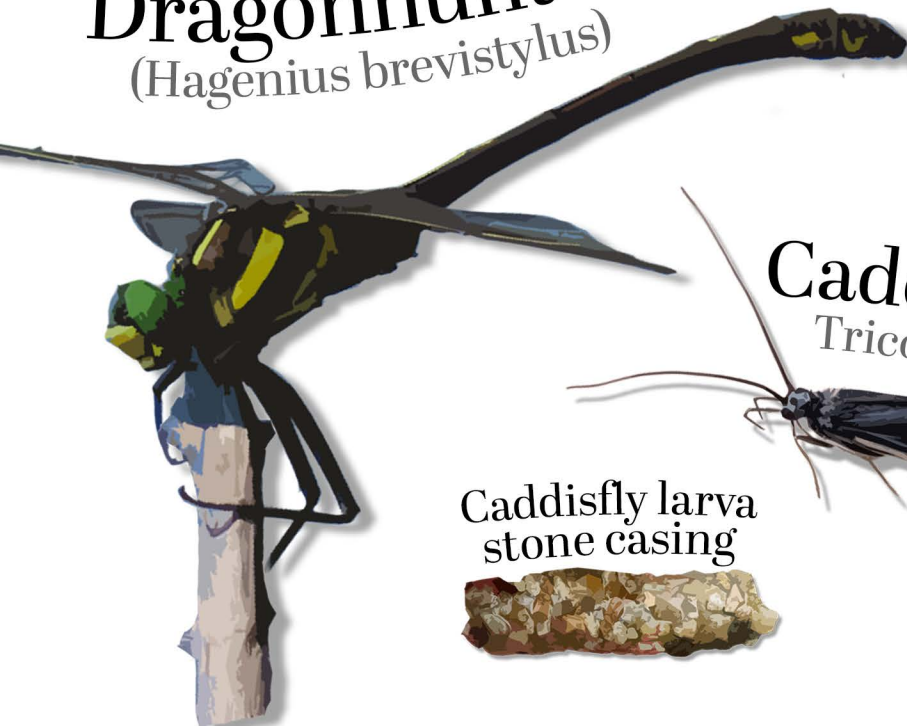
Dragonhunter
larva



European Mantis
(*Mantis religiosa*)



Dragonhunter
(*Hagenius brevistylus*)



Caddisfly
Tricoptera



Caddisfly larva
stone casing



Bumblebee
Bombus



BUBBLING SPRINGS

A cross-sectional diagram of a landscape with green trees on brown soil. Below the surface, a blue layer represents groundwater. Two white lines representing underground conduits cross each other in the center. Small white circles representing bubbles are shown rising from the intersection point towards the surface, where they emerge as a spring.

THE MUD PIT

A cross-sectional diagram of a landscape with green trees on brown soil. Below the surface, a blue layer represents groundwater. A small, dark, irregularly shaped pit is shown in the center of the groundwater layer, with small white circles representing bubbles rising from it.

THE CLAY PIT

A cross-sectional diagram of a landscape with green trees on brown soil. Below the surface, a blue layer represents groundwater. A small, dark, irregularly shaped pit is shown in the center of the groundwater layer, with small white circles representing bubbles rising from it.

BUBBLING SPRINGS

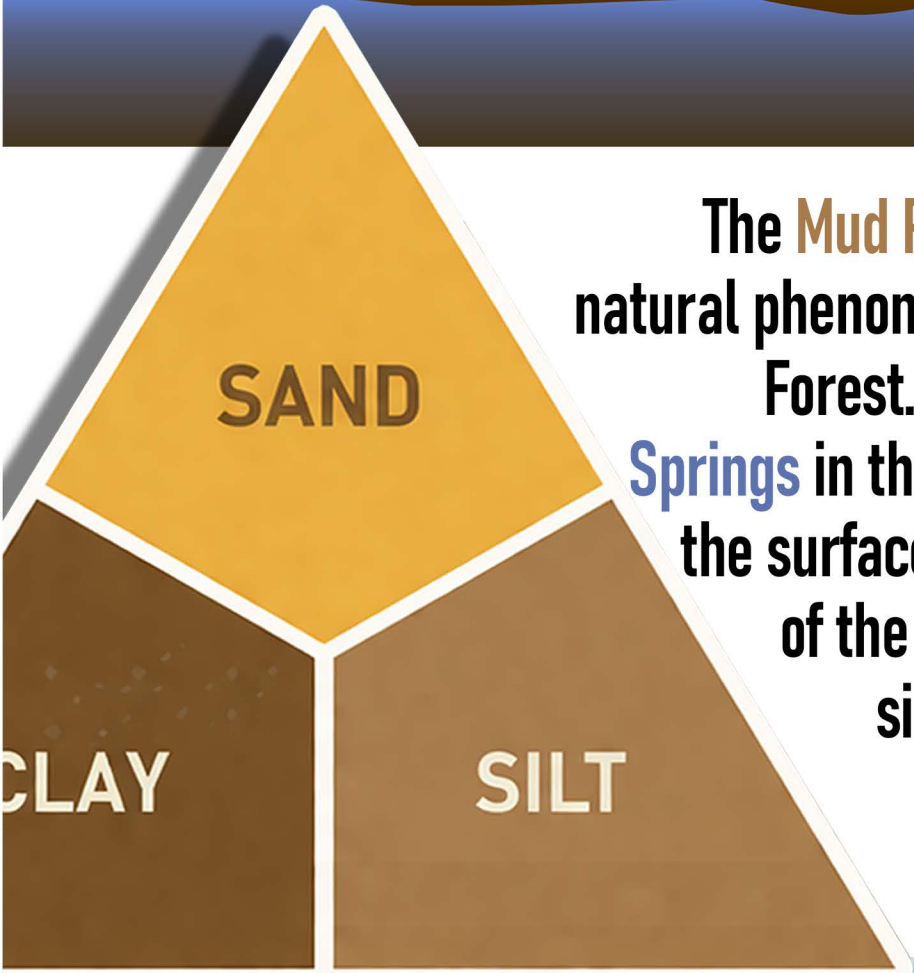


Bubbling Springs is a natural spring located in the **Wetland Forest**. Throughout the summer, it bubbles, releasing groundwater from below. Melted snow and rain collect below the dirt in the water table. Pressure from water flowing into the water table drives water through the opening at the bottom of **Bubbling Springs**, creating this magical camp phenomenon.



An illustration of a landscape with a light blue sky, green trees, and brown soil. A body of water is in the foreground, and a small stream flows through the center. The title 'THE MUD PIT' is written in large, bold, black letters across the top.

THE MUD PIT

A large, light brown triangle with a white border, divided into three smaller triangles. The top triangle is labeled 'SAND' in bold black letters. The bottom-left triangle is labeled 'CLAY' in bold black letters. The bottom-right triangle is labeled 'SILT' in bold black letters.

SAND

CLAY

SILT

The **Mud Pit** and **Clay Pit** are two more natural phenomenon located in the Wetland Forest. They are similar to **Bubbling Springs** in the way ground water reaches the surface. They are different because of the soil! Soil is made up of sand, silt, and clay, and the different soil types create the three different AGQ features

An illustration of a landscape with a light blue sky, green trees, and brown soil. A body of water is in the foreground, and a small stream flows through the center. The title 'THE CLAY PIT' is written in large, bold, black letters across the top.

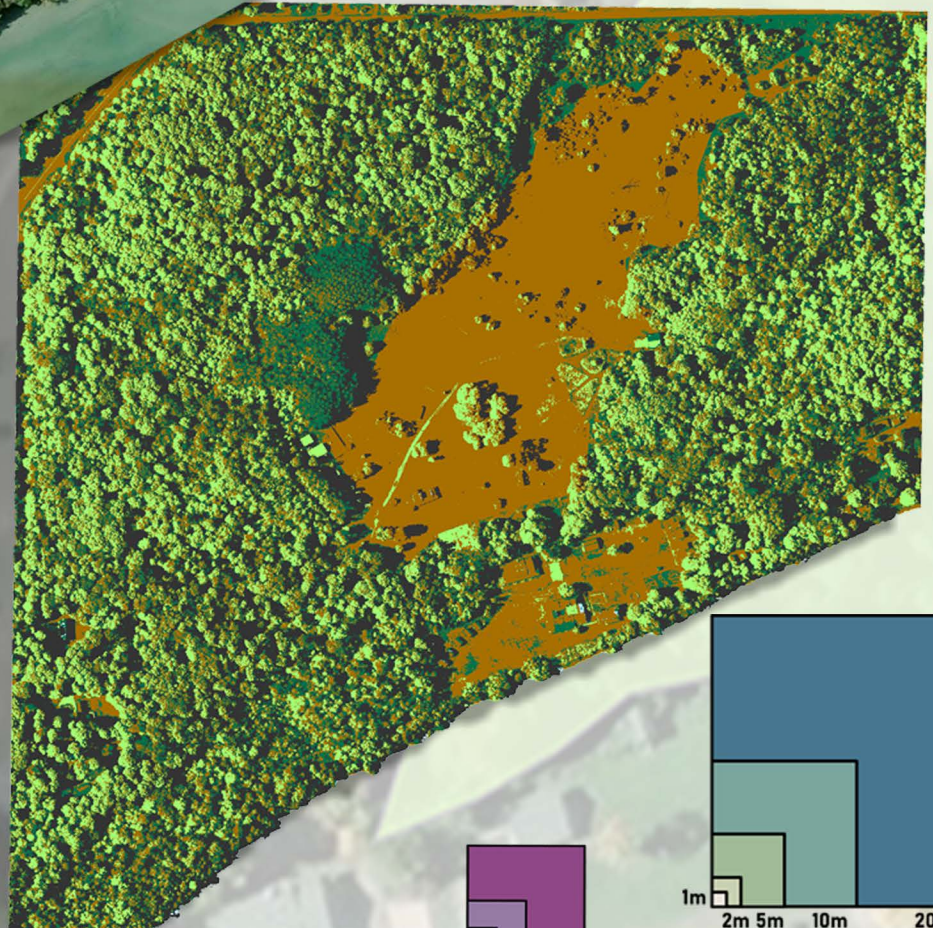
THE CLAY PIT

CARBON CAPTURE ANALYSIS

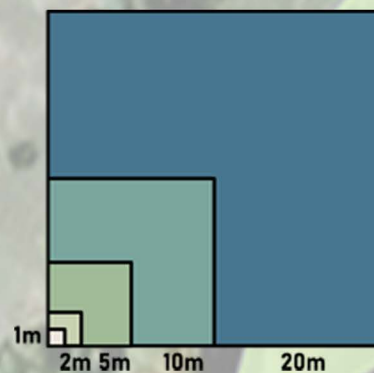
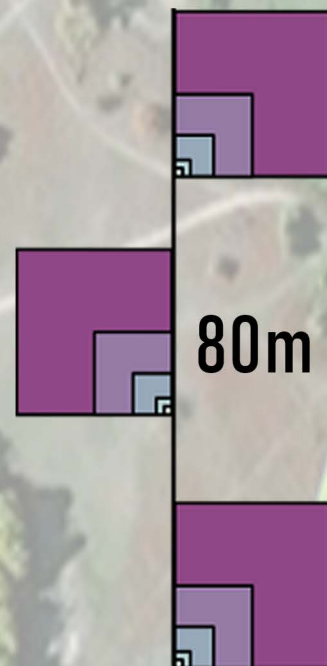


Trees and plants take carbon out of the atmosphere in order to build their leaves, bark, and all other parts. By measuring the amount of forested areas, we can estimate the amount of carbon these plants are taking out of the air.

Most of the carbon in the air is produced by humans when we drive our cars, fly in planes, grow food, and many other day-to-day activities. Carbon in the air leads to climate change. By measuring carbon capture, we can learn how Camp AGQ contributes to fighting climate change.



NESTED PLOT CARBON ESTIMATE



Calculate forested areas by digitizing satellite imagery

Name	Area (sq m)	Acres
Wetland Forest	70518	17
Aspen Maple Forest	94010	23
Hemlock Pine Forest	72529	18
Mixed Hardwood Forest	82796	20



Of forest removes **1.6** metric tons of carbon per year

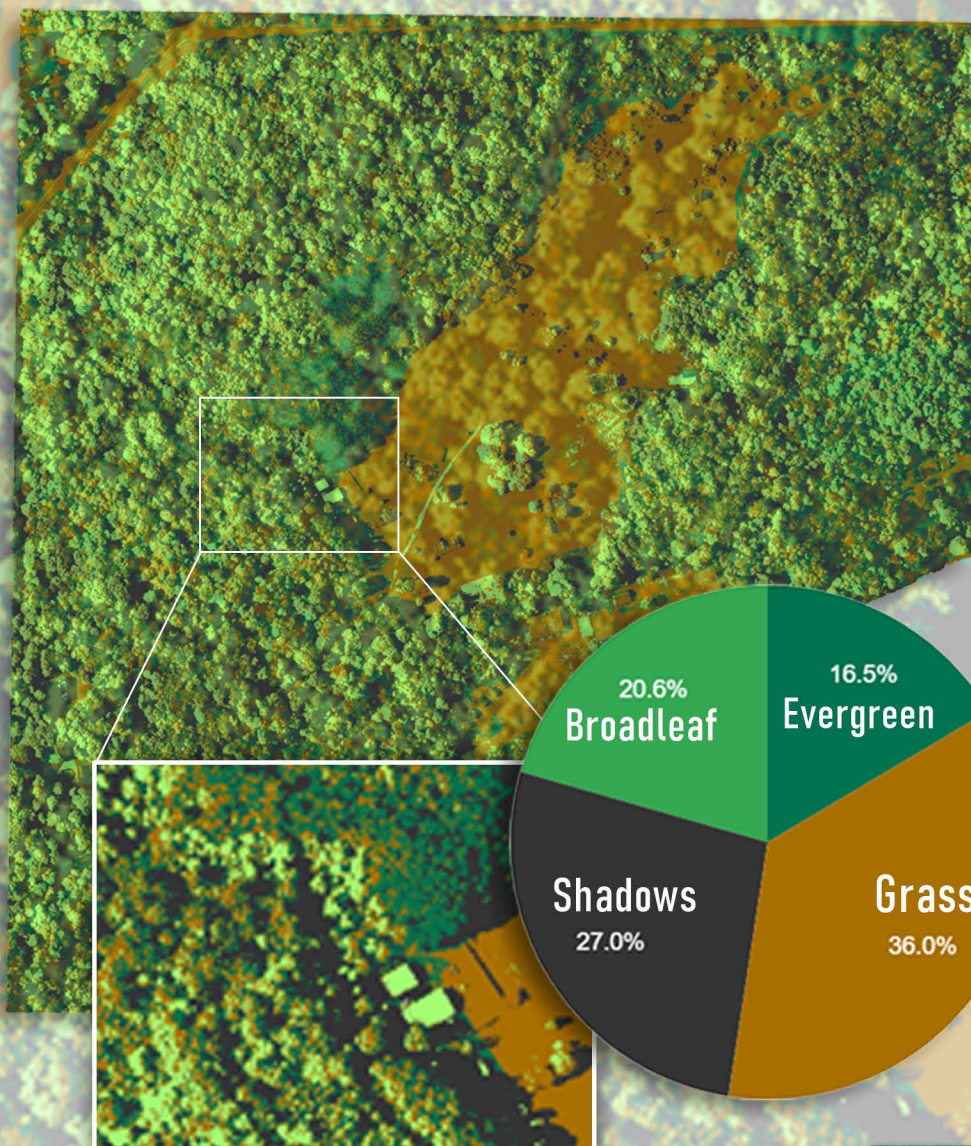
Remove **126** metric tons of carbon per year
That's like taking **26** cars off the road each year!

Use nested plots along a transect to verify forest composition

Three plots were placed along a 80 m transect and species were recorded within each plot. Nested plots let you sample the smallest species in the smallest plots and subsequently larger size classes as you work up in plot size. When looking at the data collected for a single plot, you get a glimpse at the overall habitat composition for representational areas of the forest.

IMAGE CLASSIFICATION

CARBON ESTIMATE



An image classification uses software to separate a picture into various categories of pixel. This software used 4 categories, broadleaf, evergreen, shadow, and grass and counted the number of pixels for each. The number of pixels can tell us how large an area is when you know how big an area the pixel represents

Pixel Type	Count	Area (Sq m)	Acres
Broadleaf	1039538	93558.4	23.1
Grass	2271585	204442.7	50.5
Shadow	1703879	153349.1	37.9
Evergreen	1298660	116880.2	28.9

1 ACRE

Of forest removes 1.6 metric tons of carbon per year

Each pixel is 30cm x 30 cm



Using the number of pixels, we can calculate the area of a given pixel type.

52
FORESTED
ACRES

Remove 83 metric tons of carbon per year
That's like taking 18 cars off the road each year!

CAMP AGQ MAPS



B u r t L a k e

CAMP AGQ



HEMLOCK
PINE FOREST

ASPEN MAPLE
FOREST

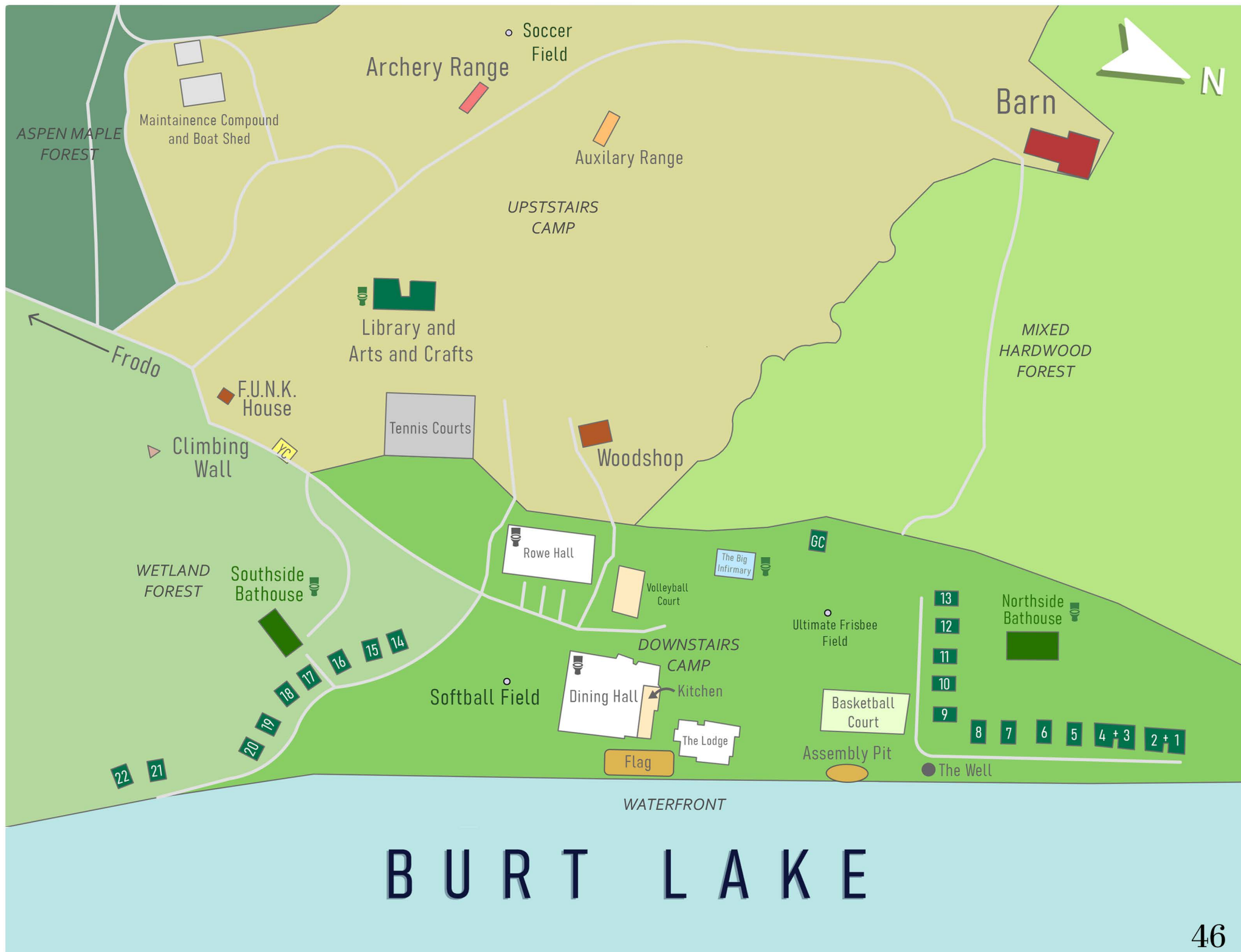
UPSTAIRS
CAMP

MIXED
HARDWOOD
FOREST

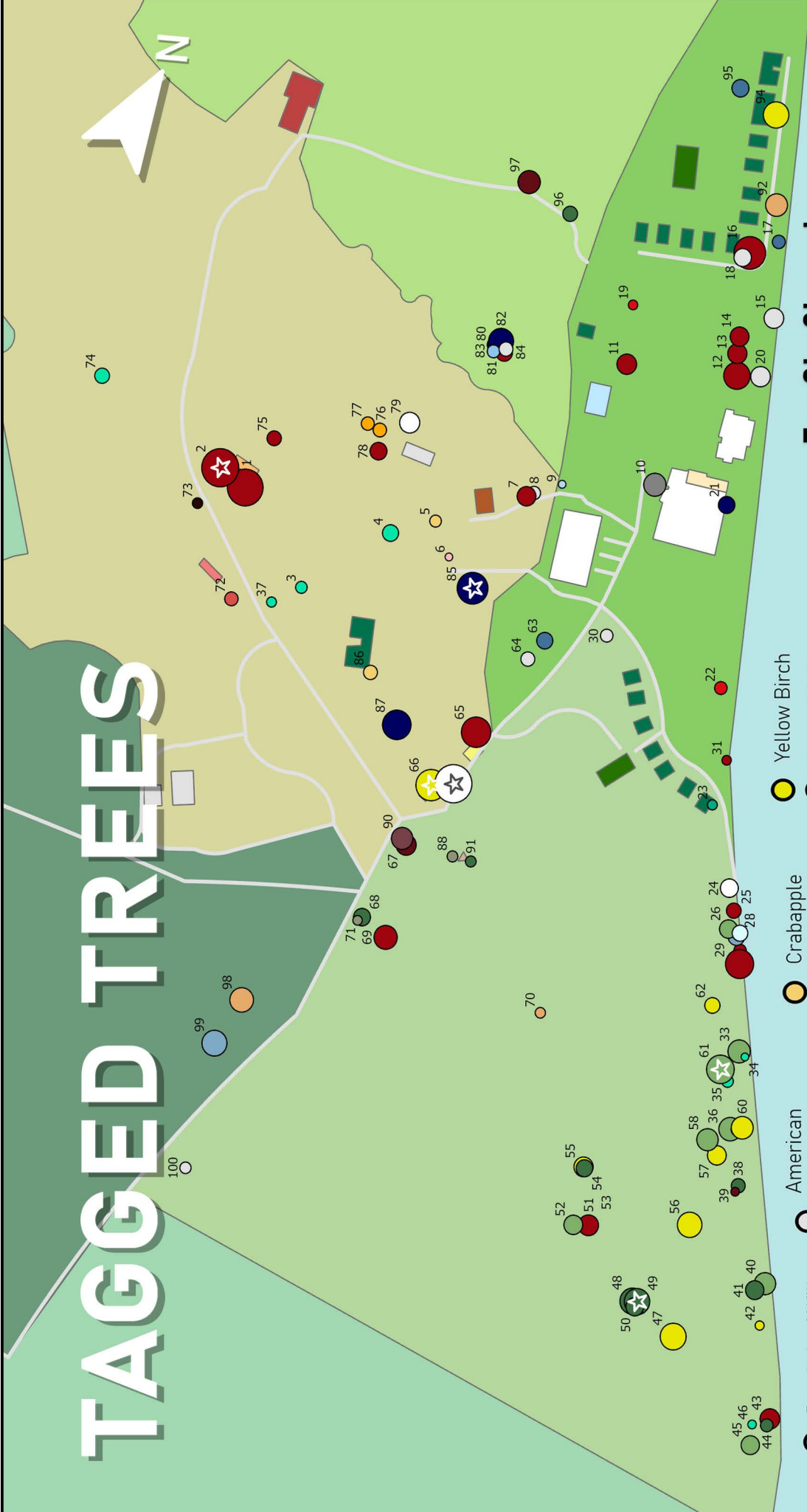
WETLAND
FOREST

DOWNSTAIRS
CAMP

WATERFRONT



TAGGED TREES



- | | | | |
|---------------------|---------------------|----------------------|---------------------|
| ○ Peach-leaf Willow | ○ American Basswood | ● Crabapple | ● Yellow Birch |
| ● Norway Maple | ● White Ash | ● Fire Cherry | ○ Paper Birch |
| ● Red Maple | ● Red/Green Ash | ○ Balsam Poplar | ● Ironwood |
| ● Sugar Maple | ● White Pine | ● Eastern Cottonwood | ● White-cedar |
| ● Mountain Maple | ● Scotch/Scots Pine | ● Trembling Aspen | ● American Beech |
| ● Staghorn Sumac | ● Eastern Hemlock | ● Bigtooth Aspen | ● Pin Oak |
| | | | ● Shagabark Hickory |

Top Six Champions

38.5in

33.7in

30.7in

39in

48.6in

49in

Champion Trees represent the largest trees within a given species. The six largest AGQ trees are shown to the right and represented with a star on the map. Icon size represents the diameter of the trunk.





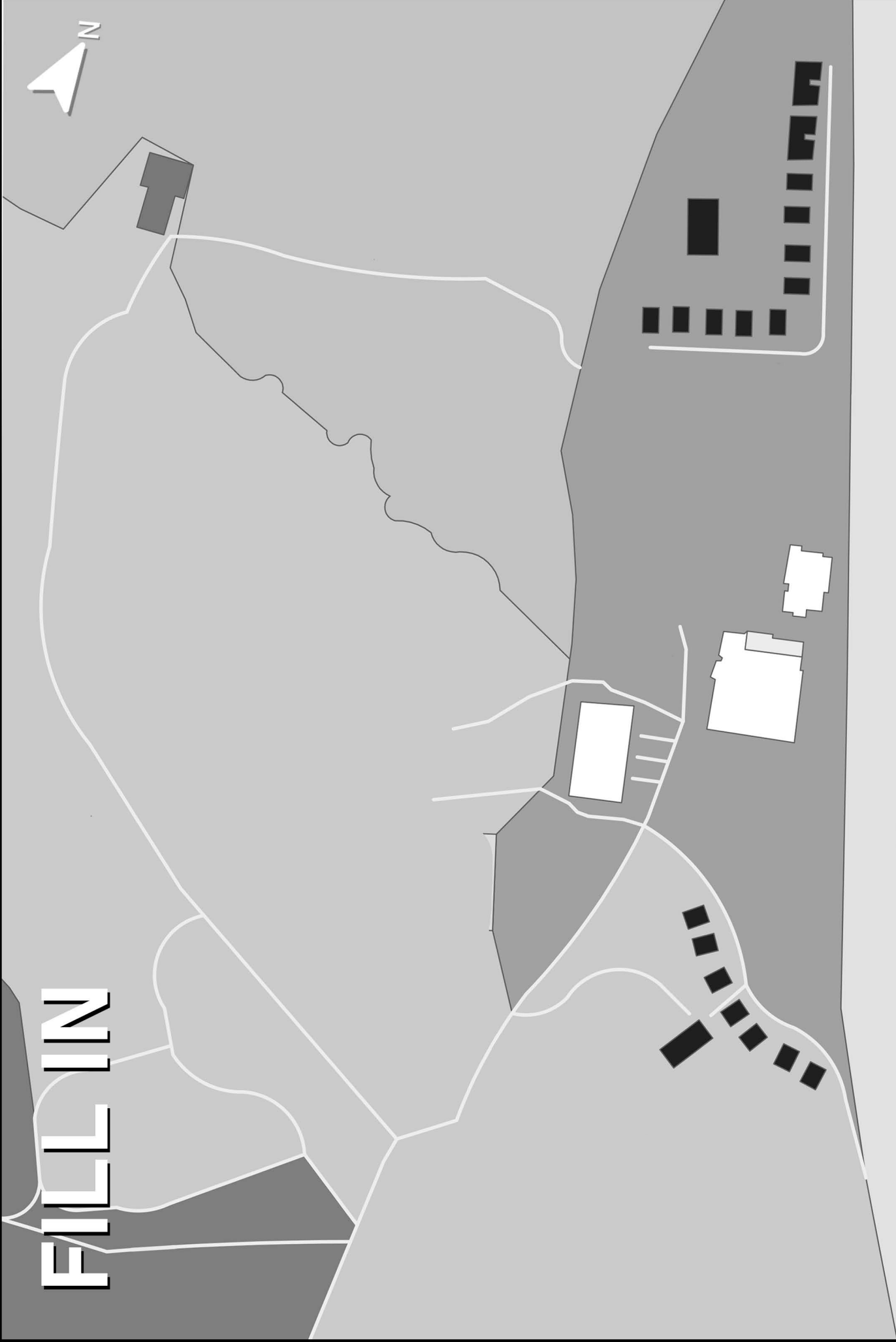
756ft

676ft

595ft

ELEVATION

FILL IN



Fill out this map as you explore Camp AGQ!

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